Stardust Sample Return Capsule Entry Trajectory Simulation and Validation against Flight Reconstruction Data

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Abstract

A MATLAB-based point-mass entry trajectory simulation was developed and validated for the Stardust sample return capsule, which reentered Earth's atmosphere on January 15, 2006. The simulation begins from a planar three-degree-of-freedom baseline using Modified Newtonian impact theory with equilibrium real-gas corrections (EquiFlow), Sutton-Graves convective heating, and an exponential atmosphere, then extends through three major and four minor upgrades, including non-planar rotating-Earth dynamics with a six-state translational formulation, Fay-Riddell stagnation-point heating using equilibrium boundary-layer edge and wall properties, free-molecular aerodynamics bridged to the continuum solution via Knudsen number, a layered 1976/1962 U.S. Standard Atmosphere, J2 oblate-Earth gravity, Tauber-Sutton radiative heating validated independently against Earth and Mars flight data, and a 2000-trial Monte Carlo landing dispersion analysis. Each component was verified independently against a closed-form solution, published correlation, or flight data prior to integration into the full model. Benchmarked against the Desai and Qualls (2010) Stardust Best Estimated Trajectory data, the baseline model predicted peak convective heating of 788.0 W/cm² (within 4% of flight data) and drogue-deployment timing within 9 percent, while over-predicting peak deceleration of 32.89 g by 10.6 to 22.5% depending upon the configuration. This bias was traced to Modified Newtonian drag over-prediction and exponential-atmosphere density over-prediction near peak-loading altitude, isolated using a layered atmosphere substitution that recovered roughly half the discrepancy. Fay-Riddell heating exceeded Sutton-Graves by a factor of 5.5 at matched trajectory points due to a low equilibrium edge Prandtl number of below 0.05, which is a characteristic of dissociated shock-layer chemistry. Monte Carlo dispersion produced a 29.80 by 5.59 km 1-sigma landing ellipse consistent with a ballistic entry signature.

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Nomenclature

Symbol		Definition
a	=	Tauber-Sutton radiative exponent
A	=	reference area, m^2
b	=	Tauber-Sutton radiative density exponent
C	=	Tauber-Sutton radiative correlation constant
C_D	=	drag coefficient
$C_{D,FM}$	=	free-molecular drag coefficient
$C_{p,e}$	=	edge specific heat, J/(kg·K)
$C_{p,max}$	=	maximum pressure coefficient (Modified Newtonian theory)
d	=	molecular collision diameter, m
D	=	auxiliary deflection angle (local vertical from geocentric radius), deg
$f(V)$	=	tabulated velocity function, Tauber-Sutton correlation
g	=	local gravitational acceleration, m/s^2
g_0	=	sea-level gravitational acceleration, m/s^2
h	=	altitude above spherical Earth, km
$h_{0,e}$	=	edge stagnation enthalpy, J/kg
h_2	=	post-shock static enthalpy, J/kg
h_w	=	wall enthalpy, J/kg
H	=	atmospheric scale height, km; also geopotential altitude, m (Eq. 22, context-dependent)
J_2	=	Earth's second zonal harmonic (oblateness coefficient)
k	=	Sutton-Graves proportionality constant
k_e	=	edge thermal conductivity, W/(m·K)
K	=	Earth's flattening coefficient
Kn	=	Knudsen number
L	=	vehicle characteristic diameter, m
L/D	=	lift-to-drag ratio
L_i	=	atmospheric layer lapse rate, K/m
m	=	vehicle mass, kg
M	=	Mach number
MW_i	=	layer-specific mean molecular weight, kg/mol
N	=	number of Monte Carlo trials
P	=	freestream static pressure, Pa
P_e	=	boundary-layer edge stagnation pressure, Pa
Pr_e	=	edge Prandtl number

$\dot{q}_c$	=	convective stagnation-point heat flux, W/cm²
$\dot{q}_{FR}$	=	Fay-Riddell stagnation-point heat flux, W/cm²
$\dot{q}_r$	=	radiative stagnation-point heat flux, W/cm²
r	=	radial distance from Earth's center, $R_e + h$, km
r_b	=	vehicle base radius, m
r_{eq}	=	Earth's equatorial radius, km
r_n	=	nose radius, m
R	=	specific gas constant, J/(kg·K)
R_e	=	spherical Earth radius, km
R_i	=	layer-specific effective gas constant, J/(kg·K)
R_0	=	ellipticity-corrected local radius, km
$\bar{R}$	=	universal gas constant, J/(mol·K)
t	=	time from entry interface, s
T	=	freestream static temperature, K
T_w	=	wall temperature, K
u_e	=	boundary-layer edge velocity, m/s
V	=	inertial speed, km/s
V_2	=	post-shock velocity, m/s
V_∞	=	freestream velocity, m/s
x_i	=	perturbed Monte Carlo input parameter
Δ_{cr}	=	crossrange distance relative to nominal landing point, km
Δ_{dr}	=	downrange distance relative to nominal landing point, km

Greek

Symbol		**Definition**
β	=	ballistic coefficient, kg/m²
γ	=	flight path angle (positive below local horizontal), deg
δ_c	=	cone half-angle, deg
δ_i	=	Monte Carlo perturbation, Gaussian, truncated $\pm 3\sigma$
ϵ	=	density ratio across bow shock, ρ_∞ / ρ_2
θ	=	longitude, deg
λ	=	mean free path, m
μ	=	Earth's gravitational parameter, m³/s²
μ_e, μ_w	=	edge, wall dynamic viscosity, Pa·s
ρ	=	freestream atmospheric density, kg/m³

Symbol		Definition
ρ_0	=	sea-level (or entry-interface reference) density, kg/m³
ρ_e, ρ_w	=	edge, wall density, kg/m³
σ	=	bank angle, deg
σ_i	=	standard deviation of Monte Carlo input parameter i
Σ	=	sample covariance matrix of landing-point distribution, km²
ϕ	=	latitude, deg; also bridging-function angle, Eq. (21) (context-dependent)
ψ	=	heading angle, deg
ω	=	Earth's rotation rate, rad/s

Subscripts

Symbol		Definition
c	=	continuum
e	=	boundary-layer edge
FM	=	free-molecular
nom	=	nominal (unperturbed) condition
w	=	wall
∞	=	freestream

1. Introduction

Planetary entry, descent, and landing (EDL) remains one of the most demanding phases of any mission that returns a vehicle through a planetary atmosphere. A successful EDL design must simultaneously satisfy tight constraints on structural loading, thermal protection, and landing accuracy, while the vehicle transits flow regimes ranging from free-molecular to continuum, and from hypersonic to subsonic. Because full-scale flight testing is costly and rare, EDL vehicle design often relies heavily on trajectory simulation validated against the limited set of missions for which detailed flight data exists. Similarly, the Stardust mission provides an unusually demanding and well-documented test case for such validation.

Launched in 1999 to collect samples from comet Wild 2, Stardust's sample return capsule reentered Earth's atmosphere on January 15, 2006 at approximately 12.8 km/s (the highest velocity of any human-made object to return to Earth) [1]. At these speeds, radiative heating becomes comparable to convective heating at the stagnation point. Stardust's ballistic, spin-stabilized, 60° sphere-cone geometry, along with the detailed post-flight Best Estimated Trajectory (BET) reconstruction, makes it a rigorous and well-constrained benchmark against which a trajectory simulation's aerodynamic, aerothermal, and dynamic modeling assumptions can each be independently checked or verified [1].

This paper presents a MATLAB-based point-mass entry trajectory simulation for the Stardust vehicle, beginning from a baseline 3-degree-of-freedom (DOF) planar model and extending it through a series of major and minor upgrades addressing specific aspects of entry physics, including rotating-Earth non-planar dynamics, independent convective and radiative heating correlations, continuum-to-free-molecular aerodynamic bridging, and Monte Carlo dispersion analysis. By investigating the influence of these upgrades on a Stardust entry trajectory, this study aims to contribute a rigorously verified and extensible framework for an EDL simulation.

In what follows, the governing equations and modeling assumptions for the baseline planar trajectory and for each major and minor upgrade are outlined in Section 2. Following this, Section 3 presents the results of the baseline model against the Stardust's BET, along with the results from the aforementioned major and minor upgrades. Section 4 includes a detailed discussion of these results. Finally, Section 5 encapsulates the conclusions drawn from this work, highlighting the key insights and potential implications for future entry trajectory simulation efforts.

2. Methodology

2.1. Baseline Planar Trajectory Model

The baseline simulation integrates a planar three-degree-of-freedom point-mass model with state vector $[V, \gamma, h]$. The governing equations are:

$$\dot{V} = -\frac{\rho V^2}{2\beta} + g\sin\gamma \tag{1}$$

$$\dot{\gamma} = -\frac{V\cos\gamma}{R_e + h} + \frac{g\cos\gamma}{V} - \frac{\rho V}{2\beta}\left(\frac{L}{D}\right)\cos\sigma \tag{2}$$

$$\dot{h} = -V\sin\gamma \tag{3}$$

Stardust flew a ballistic, spin-stabilized trajectory (nominal $L/D = 0$), so the bank-angle term in Eq. (2) is inactive in the baseline case. Freestream density follows an exponential atmosphere,

$$\rho(h) = \rho_0 e^{-\frac{h}{H}} \tag{4}$$

and gravity follows an inverse-square model:

$$g(h) = g_0 \left(\frac{R_e}{R_e + h}\right)^2 \tag{5}$$

The axial force coefficient is computed from Modified Newtonian (MN) impact theory,

$$C_{p,max} = 2 - \epsilon \tag{6}$$

integrated over Stardust's spherically blunted 60° cone geometry as a nose-cap and frustum contribution weighted by the ratio of nose radius to base radius. Because Stardust's entry velocity drives post-shock temperatures well into the dissociation and ionization regime for much of the trajectory, ϵ is obtained from an equilibrium real-gas normal-shock solution (EquiFlow) whenever post-shock temperature exceeds 800 K, and from the calorically-perfect-gas (CPG) result otherwise, avoiding unnecessary real-gas solves where the flow is not significantly reacting. Convective stagnation-point heating is computed from the Sutton-Graves correlation, Eq. (7),

$$\dot{q}_c = k \sqrt{\frac{\rho}{r_n}}\, V^3 \tag{7}$$

Because MN theory is not representative of drag near and below the transonic regime, the drag coefficient is frozen at its value when the local Mach number first falls to 1.2, and this frozen value is carried through the remainder of the descent. The planar equations of motion (Eq. 1–3) are integrated using MATLAB's `ode45`, with integration terminated by root-finding events on ground impact ($h = 0$) or atmospheric skip-out.

2.2. Major Upgrade 1 – Non-Planar Rotating-Earth Dynamics

The planar model of Section 2.1. is extended to a non-planar, six-state formulation capable of tracking heading and ground track over a rotating Earth in the first major upgrade. The state vector is augmented to $[V,\ \gamma,\ h,\ \psi,\ \phi,\ \theta]$. The governing equations, expressed in an Earth-fixed rotating frame, are

$$\dot{V} = -\frac{\rho V^2}{2\beta} + g \sin\gamma - \omega^2 r \cos\phi(sin\,\gamma\, cos\,\phi + cos\,\gamma\, sin\,\phi\, sin\,\psi) \tag{8}$$

$$\begin{aligned}\dot{\gamma} = &-\frac{\mathrm{V}\cos\gamma}{r} + \frac{\mathrm{g}\cos\gamma}{V} - \frac{\rho V}{2\beta}\left(\frac{L}{D}\right)\cos\sigma - 2\omega\cos\phi\cos\psi \\ &-\frac{\omega^2 r}{V}\cos\phi(cos\,\gamma\, cos\,\phi - sin\,\gamma\, sin\,\phi\, sin\,\psi)\end{aligned} \tag{9}$$

$$\dot{h} = -V\sin\gamma \tag{10}$$

$$\dot{\psi} = \frac{\rho V}{2\beta}\left(\frac{L}{D}\right)\frac{\sin\sigma}{\cos\gamma} - \frac{V\cos\gamma\cos\psi\tan\phi}{r} - 2\omega(\tan\gamma\cos\phi\sin\psi + \sin\phi)$$
$$- \frac{\omega^2 r}{V\cos\gamma}\sin\phi\cos\phi\cos\psi \tag{11}$$

$$\dot{\phi} = \frac{V}{r}\cos\gamma\sin\psi \tag{12}$$

$$\dot{\theta} = \frac{V}{r}\cos\gamma\frac{cos\,\psi}{cos\,\phi} \tag{13}$$

The ω^2 terms in Eqs. (8)-(9) and (11) represent centrifugal acceleration due to Earth's rotation, and the $2\omega(\cdot)$ terms in Eqs. (9) and (11) represent Coriolis acceleration; both vanish identically for $\omega = 0$ which recovers the non-rotating case.

These equations are derived under the convention that γ is negative below the local horizontal (LH). Because this project defines γ as positive below the LH throughout as mentioned in Section 2.1., Eqs. (8)-(13) were obtained by substituting $\gamma \to -\gamma$ into the source formulation and simplifying. Equation (11) exhibits a $1/\cos\gamma$ singularity as flight path angle approaches 90°, which occurs naturally near ground impact as the vehicle approaches terminal velocity and descends nearly vertically. To avoid this singularity while retaining the trajectory through its aerodynamically and thermally significant portion, integration is terminated at $\gamma = 85°$ rather than at true ground impact. The resulting horizontal position error relative to true impact is negligible, since the vehicle's remaining horizontal travel between $\gamma = 85°$ and true impact is small by construction.

2.3. Major Upgrade 2 – Fay-Riddell Stagnation-Point Heating

The second major upgrade computes stagnation-point convective heating from the Fay-Riddell relation, Eq. (14), evaluated using equilibrium real-gas properties rather than the calorically-perfect-gas assumptions implicit in Eq. (7)'s correlation coefficients. The Fay-Riddell heat flux is

$$\dot{q}_{FR} = 0.763\, Pr_e^{-0.6}(\rho_e\mu_e)^{0.4}(\rho_w\mu_w)^{0.1}\sqrt{\frac{du_e}{dx}}\left(h_{0,e} - h_w\right) \tag{14}$$

And the stagnation-point velocity gradient is estimated from MN theory as

$$\frac{du_e}{dx} = \frac{1}{r_n}\sqrt{\frac{2(P_e - P_\infty)}{\rho_e}} \tag{15}$$

Edge conditions are obtained from a three-step equilibrium real-gas property chain, evaluated with EquiFlow at each sampled trajectory point. First, the freestream is passed through an equilibrium normal shock to obtain post-shock density, from which ϵ and $C_{p,max}$ (Eq. 6) give the edge stagnation pressure and the edge stagnation enthalpy,

$$P_e = P_\infty + C_{p,max}\left(\frac{1}{2}\rho_\infty V_\infty^2\right) \tag{16}$$

$$h_{0,e} = h_2 + \frac{1}{2}V_2^2 \tag{17}$$

Second, the full equilibrium edge state (ρ_e, μ_e, $C_{p,e}$, and k_e) is obtained from an enthalpy-pressure equilibrium solve at $(h_{0,e},\ P_e)$, from which the edge Prandtl number is computed directly,

$$Pr_e = \frac{\mu_e C_{p,e}}{k_e} \tag{18}$$

Lastly, the wall state (ρ_w, μ_w, h_w) is obtained from a temperature-pressure equilibrium solve at a fixed cold-wall temperature $T_w = 300$ K, the edge pressure P_e, assumes a Lewis number of unity, and a fully catalytic wall. Because each evaluation of Eqs. (14)-(18) requires three equilibrium chemistry solves, Fay-Riddell heating is computed at a sparse set of sample points (32 total points) spanning the region of the trajectory where Sutton-Graves heating exceeds 1% of its peak value, rather than at every integration step.

2.4. Major Upgrade 3 – Free-Molecular Aerodynamics and Continuum Bridging

The base model assumes continuum flow throughout the trajectory whereas, near the entry interface, where freestream density is low enough that the mean free path becomes comparable to the vehicle's characteristic dimension, this assumption breaks down. This upgrade computes free-molecular aerodynamic coefficients following Regan and Anandakrishnan's treatment of rarefied hypersonic flow [2] and bridges them to the continuum result of Section 2.1. as a function of local Knudsen number, evaluated for a single-species nitrogen approximation of air. Flow is continuum for $Kn < 0.001$, free-molecular for $Kn > 10$, and transitional between.

$$\lambda = \frac{kT}{\pi\sqrt{2}\, d^2 P} \tag{19}$$

Free-molecular drag coefficients for a sphere and a sharp cone under diffuse molecular reflection are taken from [2], evaluated with full thermal accommodation ($\sigma_N = \sigma_T = 1$), a cold wall ($T_w = 300$ K), and the freestream molecular speed ratio defined therein. Because these results apply to complete and standalone bodies, Stardust's blunted-cone geometry is approximated as a nose cap behaving as a complete sphere of radius and a frustum behaving as a complete cone of half-angle, combined by area fraction consistent with the continuum weighting of Section 2.1.,

$$C_{D,FM} = \left(\frac{r_n}{r_b}\right)^2 C_{D,FM,sphere} + \left[1 - \left(\frac{r_n}{r_b}\right)^2\right] C_{D,FM,cone} \tag{20}$$

The continuum result of Eq. (6) and the free-molecular result of Eq. (20) are combined via the bridging function

$$C_D = C_{D,c} + \left[C_{D,FM} - C_{D,c}\right] sin^2\phi, \qquad \phi = \pi\left(\frac{3}{8} + \frac{1}{8}\right) log_{10}(Kn) \tag{21}$$

for $0.001 \leq Kn \leq 10$, with $C_D = C_{D,c}$ below this range and $C_D = C_{D,FM}$ above it. Equation (21) is evaluated at every point of the drag-coefficient table underlying the baseline and all subsequent aerodynamic models, so continuum-regime results are unaffected wherever $Kn < 0.001$ and diverge from the baseline only where the free-molecular contribution becomes significant.

2.5. Minor Upgrade 1 – Extended Atmosphere Model

The exponential atmosphere of Eq. (4) is replaced with a layered model following the 1976 U.S. Standard Atmosphere (0-86 km) and its 1962 Standard Atmosphere extension (86-150 km), providing a more physically representative density profile across the full range of altitudes traversed during entry. Below 86 km, geometric altitude is first converted to geopotential altitude:

$$H = \frac{h * r_E}{r_E + h} \tag{22}$$

The atmosphere is divided into seven layers, each with a specified base geopotential altitude, base temperature, base pressure, and constant lapse rate, with temperature within a layer given by

$$T = T_{base} + L_i(H - H_{base}) \tag{23}$$

Pressure is obtained by integrating the hydrostatic equation through each layer in sequence, using the isobaric form for isothermal layers ($L_i = 0$) and the polytropic form otherwise,

$$P = \begin{cases} P_{base} \exp\left(\frac{g_0(H - H_{base})}{RT}\right), & L_i = 0 \\ P_{base}\left(\frac{T}{T_{base}}\right)^{-\frac{g_0}{L_i R}}, & L_i \neq 0 \end{cases} \tag{24}$$

with each layer's base pressure set equal to the top-of-layer pressure of the preceding layer, and density obtained from the ideal gas law $\rho = P/RT$. Above 86 km, where the mean molecular weight of air begins to vary with altitude, geometric rather than geopotential altitude is used directly, and each layer employs a layer-specific effective gas constant R_i reflecting the changing atmospheric composition, following the same layered lapse-rate integration structure as Eqs. (23)-(24).

2.6. Minor Upgrade 2 – J2 Oblate-Earth Gravity

The inverse-square gravity model of Eq. (5) treats Earth as a perfect sphere. Minor Upgrade 2 replaces this with a latitude-dependent gravity model accounting for Earth's oblateness (the J_2

zonal harmonic) and the centrifugal effect of Earth's rotation, where ϕ is a state variable. Local radius is first corrected for Earth's ellipticity,

$$R_0 = r_{eq}(1 - K\, sin^2\phi), R = R_0 + h \tag{25}$$

An auxiliary angle accounting for the deflection of the local vertical from the geocentric radius vector is defined as:

$$D = K\left(1 - \frac{h}{R_0}\right)\sin(2\phi) \tag{26}$$

and effective gravitational acceleration, including both the J_2 oblateness correction and the centrifugal reduction due to Earth's rotation, is

$$g(\phi, h) = \frac{\mu}{R^2}\left[1 - \frac{3}{4}J_2(1 - 3\cos{(2\phi)})\right] - R\omega^2\cos\,\phi\cos{(\phi - D)} \tag{27}$$

consistent with the rotating-frame formulation of Section 2.2. Eq. (27) replaces Eq. (5) as the gravity term in Eqs. (8)-(9) when the non-planar J2 gravity model is selected; Earth is treated as spherical and non-rotating for gravity purposes ($g = g_0$, Eq. 5) otherwise.

2.7. Minor Upgrade 3 – Tauber-Sutton Radiative Heating (Earth and Mars)

Stagnation-point radiative heating is computed using the Tauber-Sutton correlation [3], and added to convective heating (Eq. 7) as $\dot{q}_{total} = \dot{q}_c + \dot{q}_r$. The correlation has the general form:

$$\dot{q}_r = C\, r_n^a\, \rho^b\, f(V) \tag{28}$$

where $f(V)$ is a tabulated, empirically fit function of freestream velocity specific to each planet's atmospheric composition, obtained by log-space interpolation of the source's tabulated values. For Earth (air), $b = 1.22$ and the exponent a is itself a function of local velocity and density rather than a constant,

$$a = \min(1.072\, \times\, 10^6 V^{-1.88}\rho^{-0.325}, 1) \tag{29}$$

with $C = 4.736 \times 10^4$, valid for $V = 10$–16 km/s, $\rho = 6.66 \times 10^{-5}$ to 6.31×10^{-4} kg/m³, and $r_n = 0.3$–3 m. Equation (28) is evaluated only within each planet's stated velocity range and set to zero elsewhere, since radiative heating is physically negligible at lower entry speeds and the correlation is not intended to extrapolate beyond it. The Earth correlation is evaluated along the Stardust trajectory of Sections 2.1–2.4. The same correlation, evaluated with its Mars-specific coefficients and velocity function, was additionally cross-validated against independent Mars Pathfinder flight data [4] as a check on the implementation's generality across planetary atmospheres (Appendix D).

2.8. Minor Upgrade 4 – Monte Carlo Landing Dispersion Analysis

Using the non-planar formulation of Section 2.2., a Monte Carlo dispersion analysis characterizes the landing footprint of the Stardust trajectory under representative uncertainty in atmospheric density and entry conditions. Each of N trials independently perturbs atmospheric density bias, entry flight path angle, entry velocity, and initial latitude and longitude, drawn from independent Gaussian distributions truncated at $\pm 3\sigma$,

$$x_i = x_{nom} + \delta_i, \delta_i \sim \mathcal{N}(0, \sigma_i^2)\ |_{-3\sigma_i}^{+3\sigma_i} \tag{30}$$

with atmospheric density dispersed as a single multiplicative bias held constant along each trial's trajectory.

$$\rho(h) = \rho_0\left(1+\delta_\rho\right)e^{-\frac{h}{H}} \tag{31}$$

Each perturbed initial condition is propagated to ground impact using Eqs. (8)-(13), and the resulting landing latitude and longitude are converted to downrange and crossrange distance relative to the nominal landing point using a local flat-Earth approximation,

$$\Delta_{dr} = R_e(\theta - \theta_{nom}),\ \Delta_{cr} = R_e(\phi - \phi_{nom}) \tag{32}$$

The landing ellipse is characterized by the sample covariance matrix of the downrange-crossrange landing point distribution across all successful trials,

$$\Sigma = \mathrm{cov}([\Delta_{dr}, \Delta_{cr}]) \tag{33}$$

and the eigen-decomposition of Σ gives the semi-major and semi-minor axes of the 1σ landing ellipse as the square roots of the eigenvalues, with orientation given by the corresponding eigenvector.

3. Results

3.1. Baseline Model – Stardust Validation

The baseline configuration (`aero_mode='equiflow'`, `motion_mode='planar'`, `atm_model='exponential'`, `gravity_model='inverse-square'`) was integrated from entry interface to ground impact. MN drag was frozen at Mach 1.20 ($t = 127.24$ s, $h = 30.5$km, $C_D = 1.116$, $\beta = 79.4$kg/m²) and carried to impact at $t = 475.21$s ($V = 35.8$ m/s, $\gamma \rightarrow 90°$) to be consistent with the near-vertical, terminal-velocity descent expected of a ballistic entry. Figure 1 presents the full trajectory time history. Figure 1(a) shows flight path angle, velocity, and altitude versus time where flight path angle increases monotonically from the 8.21° entry value toward 90° as the vehicle approaches terminal descent, while velocity and altitude both decays rapidly over the first 100 s before flattening into the extended low-altitude, near-terminal-velocity. Figure 1(b) shows altitude and load factor versus time, with the load-factor peak occurring well before the vehicle reaches low altitude, which is consistent with peak deceleration occurring during the hypersonic portion of descent rather than near the surface.

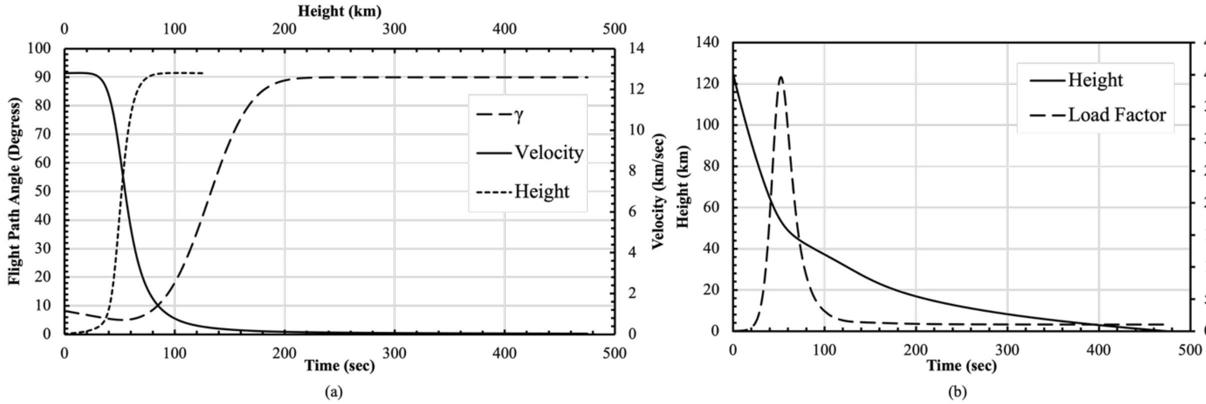

Figure 1. Baseline model Stardust entry trajectory time history: (a) flight path angle, velocity, and altitude versus time from entry interface; (b) altitude and load factor versus time from entry interface.

Figure 2 focuses on the aerothermal and drogue-relevant portion of the trajectory. Figure 2(a) compares Mach number and Sutton-Graves convective heat rate versus time from entry interface, showing the heat-rate peak trailing the Mach-number peak and gives a peak convective heating of 788.0 W/cm² at $t = 43.6$ sec ($h = 60.9$ km, $V = 10963$m/s), with an integrated convective heat load of 22,280 J/cm²**.** Figure 2(b) presents load factor together with the altitude-velocity profile, showing a peak deceleration of 39.61 g at $t = 52.2$ s ($h = 53.5$ km, $V = 8025.0$m/s) against the BET-reported peak of 32.89 g, an over-prediction of +20.4%. The same panel's altitude-velocity trace crosses the drogue-deployment reference condition ($M = 1.23$) at $t = 126.09$s and $h = 30.82$km, versus the BET's reported $t = 137.9$s and $h = 31.03$km [1]. Table 1 summarizes all validation metrics.

Table 1. Baseline model validation against Desai & Qualls (2010) [1].

Metric	Model	Reference	Error	% Error
Entry velocity*	12.799 km/s	12.799 km/s	0	0%
Entry flight path angle*	8.21°	8.21°	0	0%
Entry altitude*	124.99 km	124.99 km	0	0%
Peak deceleration	39.61 g	32.89 g	+6.72 g	+20.4%
Time at $M = 1.23$	126.09 s	137.9 s	-11.81s	-8.56%
Altitude at $M = 1.23$	30.82 km	31.03 km	-0.21km	-0.67%

**Entry velocity, flight path angle, and altitude are prescribed model inputs matched exactly to the BET's reported entry-interface conditions; they are listed for completeness and are not independent validation outcomes.*

3.2. Major Upgrade 1 – Non-Planar Rotating-Earth Dynamics

The non-planar configuration (`motion_mode='nonplanar'`, all other switches unchanged from Section 3.1.) was integrated using the rotating-Earth equations of motion (Eqs. 8-13). Similar to Section 2.2., integration is terminated at the $\gamma = 85°$ cutoff ($t = 177.76$ s, $h = 19.82$ km, $V = 153.6$ m/s) rather than at true ground impact. Figure 3(a) shows heading angle versus time. The heading remains nearly constant near its 102.9° initial value through the bulk of the trajectory ($t < 125$ s), then decreases more sharply as flight path angle steepens toward the cutoff. Because Stardust flies ballistic ($L/D = 0$), this heading change is driven entirely by the Coriolis and centrifugal terms in Eq. (11). Figure 3(b) shows the resulting ground track in downrange-crossrange coordinates, terminating at a downrange of -170.78 km and crossrange of 745.10 km relative to the entry point which is a nearly linear track and an expected result with the small, slowly varying heading change over the flight.

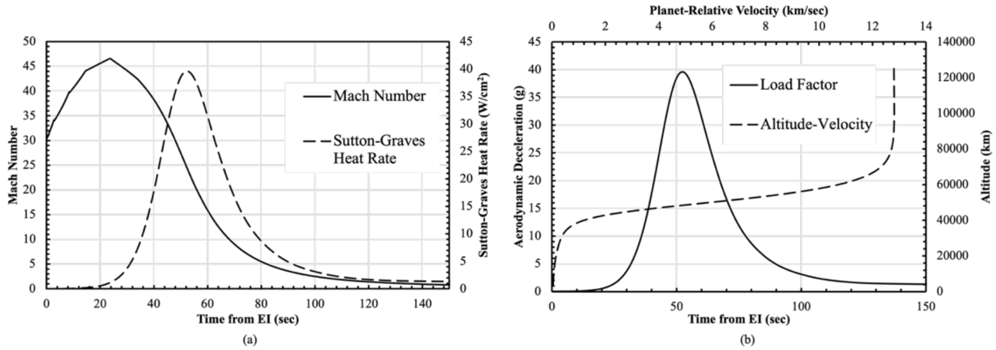

Figure 2. Baseline model aerothermal and deceleration validation: (a) Mach number and Sutton-Graves convective heat rate versus time from entry interface; (b) load factor and altitude-velocity profile versus time from entry interface, used for peak-deceleration and drogue-region Mach validation against the Best Estimated Trajectory.

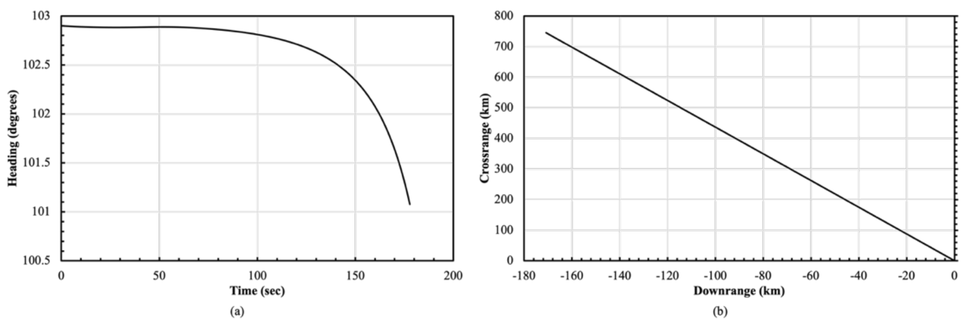

Figure 3. Non-planar trajectory results: (a) heading angle versus time from entry interface; (b) ground track in downrange-crossrange coordinates.

Table 2 compares the non-planar case against the same BET validation metrics as Table 1. Peak deceleration increased slightly to 40.29 g (vs. 39.61 g in the planar baseline, +1.7%), and the drogue-region Mach-1.23 timing shifted from -11.81s to -13.14s relative to the BET. These small differences arise because the rotating-frame ω^2 and 2ω terms in Eqs. (8)-(9) perturb translational dynamics directly, not only through the added heading/position states. Peak heating and heat load (Sutton-Graves, Tauber-Sutton, and Fay-Riddell) shift correspondingly but remain within approximately 1% of the planar baseline values reported in Section 3.1. This confirms that the added rotating-frame terms are a small perturbation on the dominant drag- and gravity-driven dynamics for this entry.

Table 2. Non-planar model validation against Desai & Qualls (2010).

Metric	Model	Reference	Error	% Error
Peak deceleration	40.29 g	32.89 g	+7.40 g	+22.5%
Time at $M = 1.23$	124.76 s	137.9 s	-13.14s	-9.53%
Altitude at $M = 1.23$	30.83 km	31.03 km	-0.20km	-0.64%

3.3. Major Upgrade 2 – Fay-Riddell Heating

Fay-Riddell heating was evaluated at 32 sampled points along the planar baseline trajectory, using the equilibrium real-gas edge and wall properties described in Section 2.3. Figure 4 compares the resulting Fay-Riddell heat rate against the Sutton-Graves convective, Tauber-Sutton radiative, and combined total heat rates over the primary heating window of the trajectory.

Fay-Riddell heating peaked at 4091.78 W/cm² at $t = 38.88$ s, more than five times the Sutton-Graves value computed at the same trajectory point (709.48 W/cm², exponential atmosphere; 749.85 W/cm² using `ATM.atm` density for a consistent freestream comparison) and well above the combined Sutton-Graves plus Tauber-Sutton peak of 953.54 W/cm² reported in Section 3.1. This large discrepancy is a direct consequence of the equilibrium edge Prandtl number computed from Eq. (18), which fell well below the value implicitly assumed by the Sutton-Graves correlation. The integrated Fay-Riddell heat load over the sampled window was 87,531 J/cm², compared to the Sutton-Graves-based total convective heat load of 22,280 J/cm².

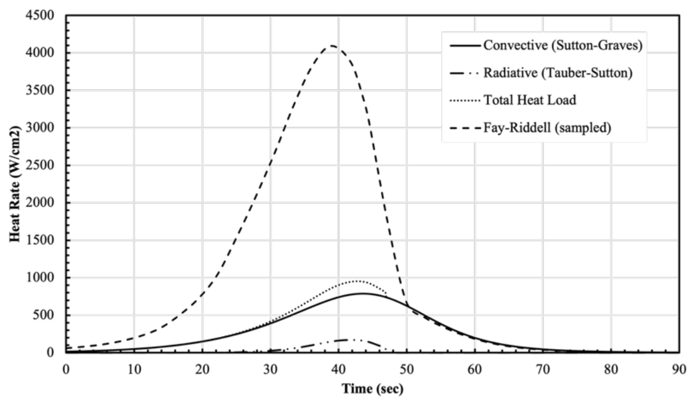

Figure 4. Comparison of Sutton-Graves convective, Tauber-Sutton radiative, combined total, and Fay-Riddell stagnation-point heat rates versus time from entry interface.

3.4. Major Upgrade 3 – Free-Molecular Aerodynamics and Bridging

Knudsen number and the resulting bridged drag coefficient were evaluated along the baseline trajectory using Eqs. (19)-(21). Figure 5(a) shows Knudsen number versus time from entry interface on a logarithmic scale. Kn increases monotonically from approximately 10^{-7} near entry interface to slightly above 10 by $h \approx 130km, t \approx 130$s and crosses the continuum threshold ($Kn = 0.001$) around $t \approx 70$-80 s and the free-molecular threshold ($Kn = 10$) only near the end of the simulated window.

Figure 5(b) compares the resulting continuum-only drag coefficient against the bridged result of Eq. (21). The bridged curve tracks the continuum value ($C_D \approx 1.39$) almost exactly through the majority of the trajectory, beginning to diverge only after $t \approx 75$ sec as Kn rises through the transitional regime, and asymptotes toward approximately 1.555 by $t \approx 120$-130 s as the free-molecular contribution becomes dominant. This behavior directly confirms the bridging function's intended limiting behavior from Section 2.4. since the model reduces to pure continuum aerodynamics wherever $Kn < 0.001$ and only meaningfully departs from it as the vehicle approaches the edge of the sensible atmosphere.

Critically, this transition occurs entirely outside the trajectory's aerodynamically and thermally significant window. Peak deceleration and peak heating both occur near $t \approx 40\text{-}52$ s (Sections 3.1. – 3.3.), well before Kn leaves the continuum regime. Consequently, incorporating free-molecular bridging produces no meaningful change to peak-g, peak heat rate, or heat load relative to the continuum-only baseline of Section 3.1. This also happens to confirm that the continuum aerodynamic modeling used throughout the rest of this work is valid for Stardust's fast, steep entry profile.

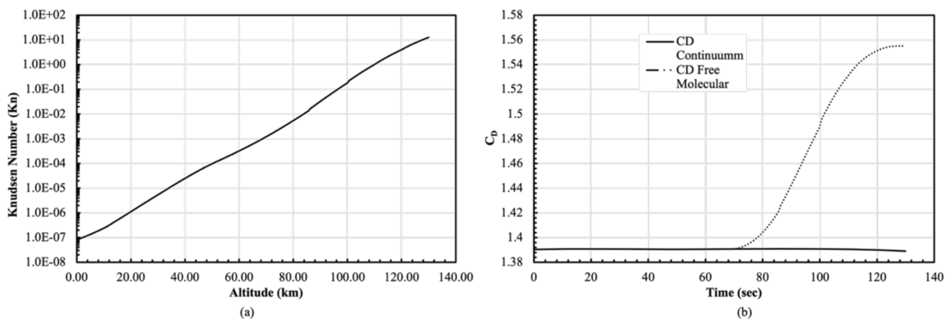

Figure 5. Free-molecular bridging results versus time from entry interface: (a) Knudsen number (log scale); (b) continuum and bridged drag coefficient.

3.5. Minor Upgrade 1 – Extended Atmosphere Model

The layered atmosphere configuration (`atm_model='layered'`, all other switches unchanged from Section 3.1.) replaces the exponential density profile with the layered 1976/1962 U.S. Standard Atmosphere model of Eqs. (22)-(24). Ground impact occurred at $t = 506.02$s, notably later than the exponential-atmosphere baseline's $t = 475.21$s. Peak deceleration was 36.38 g at $t = 51.4$s ($h = 54.2$ km, $V = 8302.6$m/s), compared to the BET-reported 32.89 g which is an over-prediction of +10.6%, less than half the +20.4% error obtained with the exponential atmosphere in Section 3.1. This improvement indicates that a meaningful portion of the baseline model's peak-deceleration bias is attributable to the exponential atmosphere's density profile near the peak-loading altitude, not solely to the MN drag coefficient identified in Section 3.1. Peak convective heating was 786.91 W/cm², and drogue-region altitude match degraded slightly, while timing agreement improved marginally. Table 3 compares the layered-atmosphere validation metrics against the exponential-atmosphere baseline of Table 1.

Table 3. Layered-atmosphere model validation against Desai & Qualls (2010), compared to the exponential-atmosphere baseline.

Metric	Layered Model	Exponential Baseline	Reference	Layered % Error
Peak deceleration	36.38 g	39.61 g	32.89 g	+10.6%
Time at $M = 1.23$	126.47 s	126.09 s	137.9 s	-8.29%
Altitude at $M = 1.23$	30.19 km	30.82 km	31.03 km	-2.70%

3.6. Minor Upgrade 2 – J2 Gravity

The J2 gravity model (Eq. 27) was first verified as a standalone component by evaluating $g(\phi, h)$ at Earth's surface ($h = 0$) for equatorial ($\phi = 0°$) and polar ($\phi = 90°$) latitude. The model returned $g_{eq} = 9.78023$ m/s² and $g_{pole} = 9.83223$ m/s², against known reference values of 9.780 m/s² and 9.832 m/s², respectively. Because these reference values are themselves measured effective-gravity values that already include the reduction due to Earth's rotation at the equator, this comparison additionally confirms that the centrifugal term in Eq. (27) is correctly incorporated and correctly signed. The J2 gravity model was then evaluated along the full non-planar Stardust trajectory (`gravity_model='j2'`). Ground impact occurred at γ-cutoff at $t = 178.04$s, with final heading, downrange, and crossrange (101.07°, -170.90 km, 745.65 km) essentially unchanged from the non-planar, inverse-square-gravity case (101.08°, -170.78 km, 745.10 km). Peak deceleration was 40.23 g, and peak convective heating was 792.76 W/cm². Table 4 summarizes the validation metrics.

Table 4. J2-gravity model validation against Desai & Qualls (2010), compared to the non-planar inverse-square baseline.

Metric	J2 Model	Non-Planar Baseline	Reference	J2 % Error
Peak deceleration	40.23 g	40.29 g	32.89 g	+22.3%
Time at $M = 1.23$	124.89 s	124.76 s	137.9 s	-9.44%
Altitude at $M = 1.23$	30.85 km	30.83 km	31.03 km	-0.59%

The negligible effect of J2 oblateness on the trajectory-level results is expected. Stardust's entry duration and the corresponding change in latitude over that time are both small relative to the spatial scale over which Earth's oblateness meaningfully varies gravitational acceleration, so this result is itself consistent with the physics rather than indicating a deficiency in the model.

3.7. Minor Upgrade 3 – Tauber-Sutton Radiative Heating

The Tauber-Sutton correlation (Eq. 28) was evaluated along the baseline Stardust trajectory. Peak radiative heating was 170.99 W/cm² at t = 41.9 s (V = 11.38 km/s), corresponding to a radiative heat load of 1956.78 J/cm². This is approximately 8.8% of the Sutton-Graves convective heat load over the same trajectory, falling just below the 9–14% range reported for Stardust in the NASA's Planetary Mission Entry Vehicles Quick Reference Guide [4]. Table 5 summarizes this result. The correlation's Mars-coefficient formulation was independently cross-validated against Mars Pathfinder entry data while reproducing published peak convective and radiative heating to within the correlation's own stated uncertainty (Appendix D).

Table 5. Tauber-Sutton radiative heating validation, Earth (Stardust).

Planet	Metric	Model	Reference	% Error
Earth	Peak radiative heat rate	170.99 W/cm²	-	-
Earth	Radiative fraction of convective load	8.8%	9-14%*	-

**Range reported for Stardust in the NASA Planetary Mission Entry Vehicles Quick Reference Guide, not a single point value.*

3.8. Minor Upgrade 4 – Monte Carlo Landing Dispersion

A 2000-trial Monte Carlo analysis was conducted using the non-planar formulation of Section 2.2. entry flight path angle, entry velocity, and initial latitude and longitude according to Eq. (30). All 2000 trials resulted in ground impact (`gamma cutoff`), with no atmospheric skip-out or other terminations observed. Figure 6 shows the resulting landing point scatter in downrange-crossrange coordinates, together with the 1σ, 2σ, and 3σ landing ellipse boundaries obtained from Eqs. (32)-(33) and the mean landing point. The mean landing point was downrange $= -171.30$ km, crossrange $= 747.17$ km relative to the nominal entry-condition landing point, with standard deviations of 8.78 km (downrange) and 29.02 km (crossrange). The resulting 1σ ellipse has a semi-major axis of 29.80 km, a semi-minor axis of 5.59 km, and is oriented at 103.4° from the downrange axis.

The ellipse's substantially larger semi-major than semi-minor axis (a ratio of approximately 5.3:1) reflects Stardust's ballistic entry. With no aerodynamic lift to redistribute dispersion across the ground track, uncertainty in entry velocity, flight path angle, and atmospheric density primarily affects how far the vehicle travels before descending (captured by the ellipse's long axis), while the ellipse's short axis is driven only by the smaller initial latitude/longitude dispersion and the trajectory's own geometric heading behavior (Section 3.2.).

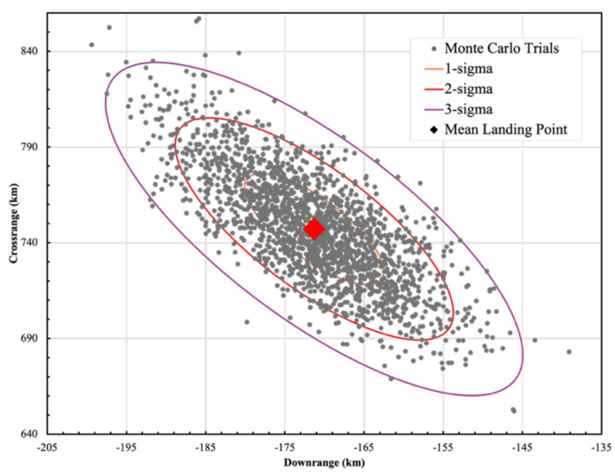

Figure 6. Monte Carlo landing point dispersion (2000 trials) in downrange-crossrange coordinates, with 1σ/2σ/3σ landing ellipse boundaries and mean landing point.

4. Discussion

4.1. Consolidated Validation Performance

Across the four configurations evaluated in Section 3 (Tables 1 – 4), peak-deceleration error was consistently over-predicted, with the non-planar and J2-gravity configurations agreeing with one another and with the planar exponential-atmosphere baseline to within approximately 2%. This result indicates that neither Earth's rotation nor its oblateness meaningfully alters the aerodynamic and thermal environment driving this entry as expected given Stardust's short entry duration and correspondingly small change in latitude over that time. The layered-atmosphere configuration, on the other hand, reduced peak-deceleration error by nearly half relative to the exponential-atmosphere baseline while leaving peak convective heating essentially unchanged as shown in Table 3. This isolates atmospheric density profile as a meaningful independent contributor to the baseline model's peak-deceleration bias, distinct from the aerodynamic modeling addressed in Section 4.2.

4.2. Systematic Peak-Deceleration Bias

The baseline model's most significant validation discrepancy is its consistent over-prediction of peak deceleration across all four configurations evaluated in Section 4.1., while independently predicting peak convective heating and drogue-region timing to within a few percent of the BET in every case. Because deceleration is directly proportional to drag coefficient at a given trajectory state, while peak convective heating depends on velocity and density through a substantially different functional form as seen in Eq. 7, the concentration of error in peak-g specifically points toward the drag coefficient itself as the primary source of bias. This is consistent with a known limitation of MN impact theory, which is derived from an inviscid, hypersonic small-disturbance argument that neglects boundary-layer displacement effects, viscous interaction, and the reduction in pressure recovery over a blunt afterbody, which, CFD solutions capture but the Newtonian pressure distribution does not. For blunt, spherically-capped bodies such as Stardust's, this typically manifests as an over-prediction of drag coefficient relative to CFD, consistent with the magnitude of deceleration error observed here.

In addition, the layered-atmosphere result in Table 3 demonstrates that this is not the sole contributor. Replacing only the atmospheric density profile, with no change to the aerodynamic model, substantially reduced peak-deceleration error. Because deceleration is directly proportional to drag coefficient at a given trajectory state, while peak convective heating depends on velocity and density through a substantially different functional form (Eq. 7), the concentration of error in peak-g specifically points toward drag coefficient as the primary source of bias. Collectively, these results suggest that the baseline model's peak-deceleration bias arises from two compounding, largely independent sources (MN drag over-prediction and exponential-atmosphere density over-prediction near peak-load altitude) rather than a single dominant cause, with the layered atmosphere model alone recovering roughly half of the total discrepancy without any change to the aerodynamic modeling itself.

4.3. Fay-Riddell / Sutton-Graves Differences and the Prandtl Number Finding

The large discrepancy between Fay-Riddell and Sutton-Graves heating at the same trajectory point as seen in Section 3.3 is directly attributable to edge Prandtl number. Because the Fay-Riddell relation (Eq. 14) scales with $Pr_e^{-0.6}$, and because Pr_e was computed directly from equilibrium real-gas properties (Eq. 18) rather than assumed at its typical air value near 0.7, this scaling is highly sensitive to how Pr_e behaves under the strongly reacting shock-layer conditions present at Stardust's entry velocity. The computed equilibrium edge Prandtl number fell well below correlation-assumed values, in some cases below 0.05. At the shock-layer temperatures present near peak heating, air is substantially dissociated and partially ionized, and equilibrium thermal conductivity in this regime is dominated by reactive and diffusive energy transport rather than simple frozen-gas conduction. Because this reactive contribution has no analog in the viscosity or specific heat terms that also appear in $Pr_e = \mu_e C_{p,e}/k_e$, the equilibrium Pr_e can depart substantially from its frozen-gas value, and a Pr_e well below unity in a strongly dissociated shock layer is a documented characteristic of equilibrium air chemistry.

Correlations such as Sutton-Graves are calibrated against a presumed, roughly constant effective Prandtl number and are not sensitive to shock-layer chemistry's actual influence on thermal transport in the way a property-resolved calculation such as Fay-Riddell is. The discrepancy observed here is therefore not evidence that either model is implemented incorrectly, but a demonstration that engineering correlations and equilibrium real-gas calculations can diverge substantially in exactly the high-enthalpy regime where accurately capturing shock-layer chemistry matters most for thermal protection system design.

4.4. Free-Molecular Bridging

Every aerodynamic and aerothermal result in this work is derived under an implicit continuum-flow assumption. Section 3.4 tested this directly by evaluating Knudsen number and a bridged drag coefficient (Eq. 21) along the same trajectory used throughout Sections 3.1–3.3. The Knudsen number remains below the continuum threshold through the entire portion of the trajectory in which peak deceleration and peak heating occur, only rising into the transitional regime well afterward, near the upper end of the descent. Because the bridged drag coefficient is, by construction, indistinguishable from the pure continuum result wherever this threshold is not exceeded, this confirms that the continuum aerodynamic and aerothermal models used throughout this work are valid over the flight regime that actually governs Stardust's structural and thermal design drivers.

Had free-molecular effects proven significant within the peak-loading or peak-heating window, every drag- and heating-dependent result in Sections 3.1–3.3 would have required re-evaluation under a corrected, blended aerodynamic model. That this did not occur is itself evidence that the baseline continuum assumption was appropriate for this entry, which was fast, steep, and rapidly transiting the thin upper atmosphere before free-molecular effects could become dynamically significant.

4.5. Known Limitations of this study

Four modeling simplifications bound the conclusions of this work and none of them materially affects the results reported here, though each defines a direction for future refinement. The layered atmosphere model (Section 2.5) exhibits an ~8% discontinuity in pressure and density at the 86 km boundary between its 1976 and 1962 constituent models which is traced to a geopotential/geometric altitude inconsistency. Because Stardust's significant trajectory segment lies well below 86 km, this does not affect any result here, but would need resolution for a slower, more gradual descent through that altitude band.

The free-molecular aerodynamic model approximates Stardust's nose cap and frustum as complete sphere and cone bodies rather than integrating pressure and shear over the vehicle's actual geometry, and both this model and the Tauber-Sutton radiative correlation are evaluated slightly below their published minimum nose-radius validity range, since Stardust's 0.22 m nose radius falls under each source's stated minimum [3]. Because free-molecular effects were shown not to influence the flight regime of interest in Section 3.4, and Tauber-Sutton agreed with flight data within its own stated uncertainty despite this extrapolation, neither is believed significant here, though both warrant attention for a smaller-nose-radius vehicle or a trajectory spending more time in the affected regimes.

The Monte Carlo analysis, as shown in Section 2.8., characterizes dispersion at ballistic ground impact using the same point-mass model as the rest of this work, not Stardust's actual parachute-assisted descent. The reported ellipse reflects free-fall dispersion alone, not the parachute-reduced footprint observed. Similarly, the non-planar formulation in Section 2.2 is a six-state, three-DOF translational model, not true 6-DOF, since attitude is not a dynamic state, and bank angle is a prescribed kinematic input rather than an integrated rotational response to aerodynamic moments. A true 6-DOF extension would require vehicle moment-of-inertia and aerodynamic moment data not available for this work.

5. Conclusion

This work developed and validated a MATLAB point-mass entry trajectory simulation for the Stardust sample return capsule, extending a planar 3-DOF baseline through three major and four minor upgrades, including non-planar rotating-Earth dynamics, Fay-Riddell real-gas heating, free-molecular aerodynamic bridging, an extended atmosphere model, J2 gravity, radiative heating, and Monte Carlo landing dispersion with each component verified independently against a closed-form solution, published reference. The baseline model matched convective heating and drogue-region timing closely with the Best Estimated Trajectory while consistently over-predicting peak deceleration. Rotating-Earth and J2 effects produced negligible additional change to peak-g and peak-heating, which was consistent with Stardust's short entry duration and correspondingly small change in latitude over that time. Fay-Riddell heating substantially exceeded Sutton-Graves at the same trajectory point because of an equilibrium edge Prandtl number well below correlation-assumed values, reflecting reactive, diffusion-dominated thermal transport in a strongly dissociated shock layer. Free-molecular bridging confirmed that continuum flow modeling remains valid throughout Stardust's aerodynamically and thermally significant trajectory segment, and the Tauber-Sutton correlation reproduced independent Earth and Mars flight data within its own stated uncertainty, which demonstrates that the implementation generalizes across planetary atmospheres. A 2000-trial Monte Carlo analysis characterized a landing footprint substantially elongated along the direction of travel, as expected of Stardust’s ballistic, lift-free entry. The limitations identified span the layered atmosphere's altitude-convention discontinuity, correlation extrapolation near Stardust's nose radius, the ballistic-only scope of the landing footprint, and the translational rather than true 6-DOF formulation. Each was shown to negligibly affect the results reported here, but each defines a clear direction for future refinement. Most notably, a true 6-DOF extension would require incorporating vehicle moment-of-inertia and aerodynamic moment data not available for this work.

Appendix

A. Layered Atmospheric Model

Table A.1. Layered Atmospheric Model Results

h (gp., m)	T_{model} (K)	T_{ref} (K)	T % err	P_{model} (Pa)	P_{ref} (Pa)	P % err	ρ_{model} (kg/m³)	ρ_{ref} (kg/m³)	ρ %err
0	288.150	288.15	0.0	101325.0	101325	0.000	1.22501	1.22500	0.001
11	216.650	216.65	0.0	22631.70	22632	-0.001	0.363916	0.36391	0.002
20	216.650	216.65	0.0	5474.72	5474.9	-0.003	0.088033	0.088035	-0.002
32	228.650	228.65	0.0	867.97	868.02	-0.005	0.013224	0.013225	-0.004
47	270.650	270.65	0.0	110.90	110.91	-0.011	0.0014274	0.0014275	-0.004
51	270.650	270.65	0.0	66.93	66.939	-0.008	0.00086155	0.00086160	-0.006
71	214.650	214.65	0.0	3.956	3.9564	-0.010	6.42×10^{-5}	6.42×10^{-5}	-0.010
86	188.889	186.87	1.08	0.2786	0.3734	-25.385	5.14×10^{-6}	6.96×10^{-6}	-26.15

B. J2 Oblate-Earth Gravity Model

Table B.1. presents the J2 gravity model's (Section 2.6.) computed surface gravity at the equatorial and polar extremes, compared against known reference values.

Table B.1. J2 gravity model verification against known equatorial and polar surface gravity values.

Latitude Case	Latitude (deg)	g_{model}(m/s²)	g_{ref}(m/s²)	% Error
Equatorial	0	9.78023	9.780	0.00236%
Polar	90	9.83223	9.832	0.00236%

Agreement is essentially exact at both latitude extremes, with error below 0.003% in each case. Because the reference values used here (9.780 m/s² equatorial, 9.832 m/s² polar) are themselves measured effective-gravity values that already incorporate the reduction due to Earth's rotation at the equator.

C. Parameter Sensitivity Study

A one-at-a-time (OAT) sensitivity study was conducted around the lifting baseline case ($\gamma_e = 8.21°$, $\beta = 61$kg/m², $L/D = 0.30$, $\sigma = 45°$, $V_e = 12{,}799$m/s), varying each parameter individually to its low and high bound while holding all others at baseline. Table C.1. summarizes the resulting outcome, peak load factor, peak stagnation heat rate, total heat load, and minimum altitude reached for all eleven cases.

Table C.1. One-at-a-time parameter sensitivity study results.

Case	γ_e (deg)	β (kg/m²)	L/D	σ (deg)	V_e (m/s)	Outcome	Peak g	Peak $\dot{q}$ (W/cm²)	Q_{total} (J/cm²)	h_{min} (km)
Baseline	8.21	61.0	0.30	45.0	12,799	Ground impact	26.88	746.82	30,424.1	−0.00
γ_e **low**	1.00	61.0	0.30	45.0	12,799	Skip-out	0.01	23.10	639.9	118.42
γ_e **high**	15.0	61.0	0.30	45.0	12,799	Ground impact	85.19	1193.21	15,802.6	−0.00
β **low**	8.21	20.0	0.30	45.0	12,799	Ground impact	29.62	443.36	16,319.5	−0.00
β **high**	8.21	200.0	0.30	45.0	12,799	Ground impact	23.83	1293.49	59,501.0	−0.00
L/D low	8.21	61.0	0.00	45.0	12,799	Ground impact	41.58	810.59	21,869.9	−0.00
L/D high	8.21	61.0	0.60	45.0	12,799	Skip-out	21.38	692.33	24,070.3	62.95
σ **low**	8.21	61.0	0.30	0.0	12,799	Skip-out	23.02	723.11	25,529.9	60.70
σ **high**	8.21	61.0	0.30	90.0	12,799	Ground impact	43.41	810.59	21,869.9	−0.00
V_e **low**	8.21	61.0	0.30	45.0	7,500	Ground impact	17.02	189.72	7,210.0	−0.00
V_e **high**	8.21	61.0	0.30	45.0	13,000	Ground impact	27.33	778.50	31,775.2	−0.00

D. Tauber-Sutton Radiative Heating: Mars Cross-Validation (Pathfinder)

As a secondary check on the Tauber-Sutton implementation, the correlation was evaluated with its Mars-specific coefficients and compared against Mars Pathfinder flight data. For Mars (assumed 97% CO_2, 3% N_2), the correlation constants in Eq. (28) are a = 0.526 and b = 1.19 (both constant, unlike the Earth case), with $C = 2.35 \times 10^4$, valid for $V = 6.5$–9 km/s, $\rho = 10^{-4}$ to 10^{-3} kg/m³, and $r_n = 1$–23 m. Because this project's baseline trajectory and atmosphere models are Earth-specific, the Mars case was evaluated separately using Mars Pathfinder's reported entry conditions, substituting Mars gravity and an exponential Mars atmosphere for Eqs. (5) and (4), respectively.

Figure D.1. shows the resulting Mars Pathfinder altitude and heat-rate time histories. Ground impact occurred at $t = 234.13$ s ($V = 170.0$ m/s). Peak convective heating (Sutton-Graves) was 96.72 W/cm² at $t = 57.6$ s ($V = 6.45$ km/s), compared to the Pathfinder's reference value of approximately 100.69 W/cm² (−3.9%). Peak radiative heating (Tauber-Sutton, Mars) was 4.31 W/cm² at $t = 53.2$ s ($V = 6.81$ km/s), compared to the reported value of approximately 5.27 W/cm² with ablation (−18.2%), within the correlation's own documented 20–30% uncertainty [4]. Integrated radiative heat load was 66.15 J/cm².

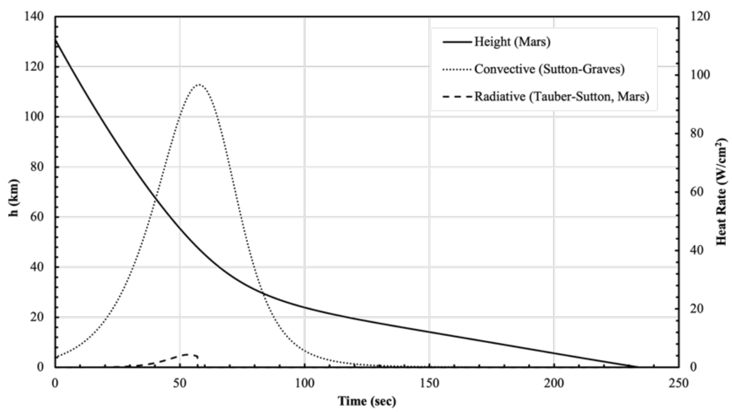

Figure D.1. Mars Pathfinder altitude and heat rates with respect to time.

Table D.1. Tauber-Sutton and Sutton-Graves validation against Mars Pathfinder.

Metric	Model	Reference	% Error
Peak convective heat rate	96.72 W/cm²	~100.69 W/cm²	−3.9%
Peak radiative heat rate	4.31 W/cm²	~5.27 W/cm²	−18.2%

Agreement with both the Earth (Stardust) and Mars (Pathfinder) flight data, obtained using the same correlation implementation with only the planet-specific coefficients and atmosphere/gravity models substituted, indicates that the Tauber-Sutton implementation is not overfit to a single planetary case and generalizes correctly across entry velocity regimes and atmospheric compositions.

E. Base Project MATLAB Script

```
%% AE 6355 - Individual Project - Stardust Entry Trajectory Simulation
% 3-DOF / 6-DOF point-mass entry simulation, validated against Stardust
% (Desai & Qualls, 2010).
%
% State (planar):    x = [V; gamma; h]
% State (nonplanar): x = [V; gamma; h; psi; phi; theta]
```

```
% gamma is positive below the local horizontal throughout.


clear; clc; close all;


%% ----------------------------------------------------------------
%% PLANET CONSTANTS AND STARDUST REFERENCE VALUES
%% ----------------------------------------------------------------
g0 = 9.81;                 % m/s^2
Re = 6378150;               % m


reference.V_EI                = 12799;
reference.gamma_EI_paper_deg  = -8.21;         % paper convention
(negative descending)
reference.r_EI                = 6503.14e3;
reference.h_EI                = reference.r_EI - Re;
reference.peak_g_BET          = 32.89;
reference.peak_g_nominal      = 32.86;
reference.drogue_time         = 137.9;
reference.drogue_h            = 31.03e3;
reference.drogue_M            = 1.23;


%% ----------------------------------------------------------------
%% USER INPUTS
%% ----------------------------------------------------------------
% aero_mode     = 'constant' | 'equiflow'
% motion_mode   = 'planar'   | 'nonplanar'
% atm_model     = 'exponential' | 'layered'
% gravity_model = 'inverse_square' | 'j2'   (j2 only active if nonplanar)
%
% Each upgrade below is demonstrated by flipping ONE switch from the
% base (equiflow / planar / exponential / inverse_square) configuration.


V0     = reference.V_EI;
gamma0 = -deg2rad(reference.gamma_EI_paper_deg);   % sign-converted to
model convention
h0     = reference.h_EI;


beta_const  = 61;           % kg/m^2 (aero_mode = 'constant' only)
LD_const    = 0;
sigma_const = 0;


aero_mode = 'equiflow';     % 'constant' or 'equiflow'
```

```
motion_mode = 'planar';     % 'planar' (3-state) or 'nonplanar' (6-state)
psi0   = deg2rad(102.9);    % rad, Stardust entry azimuth (Desai & Qualls
2010)
phi0   = 0;                 % rad, reference latitude
theta0 = 0;                 % rad, reference longitude


gravity_model = 'j2';   % 'inverse_square' or 'j2' (nonplanar only)
mu_earth    = 3.986e14;      % m^3/s^2
J2_earth    = 1.08263e-3;
omega_earth = 7.292115147e-5;% rad/s
K_ellip     = 1/298.257;


atm_model = 'layered';   % 'exponential' or 'layered'


% Stardust geometry
m_veh   = 46;
d_veh   = 0.8128;
rn      = 0.2202;
rb      = d_veh/2;
delta_c = 60*pi/180;
A_ref   = pi*rb^2;


% CPG properties / regime thresholds
gamma_air           = 1.4;
R_air               = 287;
T2_react_threshold  = 800;    % K
M_transonic_cutoff  = 1.2;


skip_altitude = h0 + 100;     % m, must exceed h0


V_grid = linspace(0, 14000, 20);
h_grid = linspace(0, 130000, 20);
assert(skip_altitude <= max(h_grid), 'skip_altitude must lie inside the
aerodynamic-table altitude range.');


cpmax_formula_version = 3;
species_model = 'earth10';


CD_table_cache_file = 'stardust_CD_table_cache.mat';
```

```
if strcmp(aero_mode, 'equiflow')
    cache_valid = false;
    if isfile(CD_table_cache_file)
        try
            loaded = load(CD_table_cache_file, 'CD_interp', 'V_grid',
'h_grid', 'rn', 'rb', 'delta_c', ...
                'gamma_air', 'R_air', 'T2_react_threshold',
'M_transonic_cutoff', ...
                'cpmax_formula_version', 'species_model', 'd_veh');
            cache_valid = isequal(loaded.V_grid, V_grid) &&
isequal(loaded.h_grid, h_grid) ...
                && loaded.rn==rn && loaded.rb==rb &&
loaded.delta_c==delta_c ...
                && loaded.gamma_air==gamma_air && loaded.R_air==R_air ...
                && loaded.T2_react_threshold==T2_react_threshold ...
                && loaded.M_transonic_cutoff==M_transonic_cutoff ...
                && loaded.cpmax_formula_version==cpmax_formula_version ...
                && loaded.d_veh==d_veh ...
                && strcmp(loaded.species_model, species_model);
        catch
            cache_valid = false;
        end
    end


    if cache_valid
        CD_interp = loaded.CD_interp;
        fprintf('Loaded cached CD(V,h) table from %s.\n',
CD_table_cache_file);
    else
        fprintf('Building CD(V,h) table...\n');
        CD_interp = build_stardust_CD_table(V_grid, h_grid, rn, rb,
delta_c, species_model, ...
            gamma_air, R_air, T2_react_threshold, M_transonic_cutoff,
d_veh);
        save(CD_table_cache_file, 'CD_interp', 'V_grid', 'h_grid', 'rn',
'rb', 'delta_c', ...
            'gamma_air', 'R_air', 'T2_react_threshold',
'M_transonic_cutoff', ...
            'cpmax_formula_version', 'species_model', 'd_veh');
    end
end


rho0 = 1.225;
H    = 7200;
```

```
%% -----------------------------------------------------------------
%% PARAMS STRUCT
%% -----------------------------------------------------------------
params.Re = Re;
switch aero_mode
    case 'constant'
        params.beta = @(V,h) beta_const;
        params.LD   = @(V,h) LD_const;
    case 'equiflow'
        params.beta = @(V,h) m_veh./(lookup_CD_checked(CD_interp, V_grid,
h_grid, V, h).*A_ref);
        params.LD   = @(V,h) 0;
    otherwise
        error('Unknown aero_mode: %s', aero_mode);
end
params.sigma   = @(t) sigma_const;
params.g_fun   = @(h) g0*(Re/(Re+max(h,0)))^2;
params.rho_fun = @(h) get_density(max(h,0), atm_model, rho0, H);
params.gravity_model = gravity_model;
params.g_j2_fun = @(phi,h) g_j2(phi, max(h,0), mu_earth, J2_earth,
omega_earth, Re, K_ellip);
params.omega = omega_earth;


LD_used = params.LD(0,0);


%% -----------------------------------------------------------------
%% ODE45 INTEGRATION
%% -----------------------------------------------------------------
switch motion_mode
    case 'planar'
        x0 = [V0; gamma0; h0];
        eom_fun = @EOM_3DOF;
        abstol_vec = [1e-3 1e-10 1e-3];
        gamma_cutoff = Inf;
    case 'nonplanar'
        x0 = [V0; gamma0; h0; psi0; phi0; theta0];
        eom_fun = @EOM_6DOF;
        abstol_vec = [1e-3 1e-10 1e-3 1e-10 1e-10 1e-10];
        gamma_cutoff = deg2rad(85);   % avoids EOM_6DOF's 1/cos(gamma)
singularity; see report Sec. 2.2.
    otherwise
        error('Unknown motion_mode: %s', motion_mode);
end
tspan = [0 2000];
```

```
freeze_time = NaN;
CD_freeze   = NaN;
beta_freeze = NaN;


ode_settings = {'RelTol',1e-8, 'AbsTol',abstol_vec, 'MaxStep',0.5};


if strcmp(aero_mode, 'equiflow')
    opts1 = odeset(ode_settings{:}, 'Events', ...
        @(t,x)
mach_ground_skip_event(t,x,gamma_air,R_air,M_transonic_cutoff,skip_altitud
e,gamma_cutoff));
    [t1, x1, te1, xe1, ie1] = ode45(@(t,x) eom_fun(t, x, params), tspan,
x0, opts1);


    if isempty(ie1)
        t = t1; x = x1; te = t1(end); xe = x1(end,:); outcome = 'time
limit';
    elseif ie1(end) == 1
        t = t1; x = x1; te = te1; xe = xe1; outcome = 'ground impact';
    elseif ie1(end) == 3
        t = t1; x = x1; te = te1; xe = xe1; outcome = 'atmospheric skip-
out';
    elseif ie1(end) == 4
        t = t1; x = x1; te = te1; xe = xe1; outcome = 'gamma cutoff';
    else
        V_freeze = xe1(1);
        h_freeze = xe1(3);
        CD_freeze   = lookup_CD_checked(CD_interp, V_grid, h_grid,
V_freeze, h_freeze);
        beta_freeze = m_veh/(CD_freeze*A_ref);
        freeze_time = te1;
        fprintf('Mach %.2f crossed at t=%.2f s (h=%.1f km, V=%.1f m/s) -
freezing CD=%.3f (beta=%.1f kg/m^2) to ground.\n', ...
            M_transonic_cutoff, te1, h_freeze/1000, V_freeze, CD_freeze,
beta_freeze);


        params2 = params;
        params2.beta = @(V,h) beta_freeze;


        opts2 = odeset(ode_settings{:}, 'Events', @(t,x)
ground_skip_event(t,x,skip_altitude,gamma_cutoff));
```

```
        [t2, x2, te2, xe2, ie2] = ode45(@(t,x) eom_fun(t, x, params2),
[te1 tspan(2)], xe1, opts2);


        t = [t1; t2(2:end)];
        x = [x1; x2(2:end,:)];


        if isempty(ie2)
            te = t2(end); xe = x2(end,:); outcome = 'time limit';
        elseif ie2(end) == 1
            te = te2; xe = xe2; outcome = 'ground impact';
        elseif ie2(end) == 3
            te = te2; xe = xe2; outcome = 'gamma cutoff';
        else
            te = te2; xe = xe2; outcome = 'atmospheric skip-out';
        end
    end
else
    opts = odeset(ode_settings{:}, 'Events', @(t,x)
ground_skip_event(t,x,skip_altitude,gamma_cutoff));
    [t, x, te, xe, ie] = ode45(@(t,x) eom_fun(t, x, params), tspan, x0,
opts);
    if isempty(ie)
        te = t(end); xe = x(end,:); outcome = 'time limit';
    elseif ie(end) == 1
        outcome = 'ground impact';
    elseif ie(end) == 3
        outcome = 'gamma cutoff';
    else
        outcome = 'atmospheric skip-out';
    end
end


fprintf('Outcome: %s at t=%.2f s (V=%.1f m/s, gamma=%.2f deg, h=%.2f
km)\n', ...
    outcome, te, xe(1), xe(2)*180/pi, xe(3)/1000);


V     = x(:,1);
gamma = x(:,2);
h     = x(:,3);


if strcmp(motion_mode, 'nonplanar')
    psi   = x(:,4);
    phi   = x(:,5);
```

```
    theta = x(:,6);
end


%% ------------------------------------------------------------
%% POST-HOC: HEATING, DECELERATION
%% ------------------------------------------------------------
rho_t = arrayfun(@(hh) get_density(max(hh,0), atm_model, rho0, H), h);


switch aero_mode
    case 'constant'
        beta_t = beta_const*ones(size(t));
    case 'equiflow'
        beta_t = nan(size(t));
        if ~isnan(freeze_time)
            pre_freeze = t <= freeze_time + 1e-9;
            beta_t(pre_freeze)  = m_veh./(arrayfun(@(vv,hh)
lookup_CD_checked(CD_interp, V_grid, h_grid, vv, hh), ...
                V(pre_freeze), h(pre_freeze)) * A_ref);
            beta_t(~pre_freeze) = beta_freeze;
        else
            beta_t = m_veh./(arrayfun(@(vv,hh)
lookup_CD_checked(CD_interp, V_grid, h_grid, vv, hh), V, h) * A_ref);
        end
end


% Sutton-Graves convective heating (report Eq. 7)
k_SG = 1.7415e-4;   % SI -> divide by 1e4 for W/cm^2
qdot_s  = k_SG*sqrt(rho_t/rn).*V.^3 / 1e4;
Q_total = trapz(t, qdot_s);
[qdot_peak, idx_heat] = max(qdot_s);


% Tauber-Sutton radiative heating, Earth (Eq. 28-29)
C_E = 4.736e4;
b_E = 1.22;
a_earth = @(Vq,rhoq) min(1.072e6 * Vq.^(-1.88) .* rhoq.^(-0.325), 1.0);


V_kms = V/1000;
in_range_TS = V_kms >= 10 & V_kms <= 16;
qdot_rad = zeros(size(V));
a_vals = a_earth(V(in_range_TS), rho_t(in_range_TS));
qdot_rad(in_range_TS) = C_E * rn.^a_vals .* rho_t(in_range_TS).^b_E ...
    .* tauber_sutton_fV(V(in_range_TS), 'earth');   % W/cm^2
Q_rad_total = trapz(t, qdot_rad);
```

```
[qdot_rad_peak, idx_rad_peak] = max(qdot_rad);
qdot_total = qdot_s + qdot_rad;


% Fay-Riddell heating (report Eq. 14-18), sampled sparsely due to cost
% of the three equilibrium chemistry solves per point.
if strcmp(aero_mode, 'equiflow')
    n_FR_samples = 31;
    heating_region = find(qdot_s >= 0.01*qdot_peak);


    if isempty(heating_region)
        error('No heating region found for Fay-Riddell sampling.');
    end


    sample_idx = round(linspace(heating_region(1), heating_region(end),
n_FR_samples));
    sample_idx = unique([sample_idx(:); idx_heat]);
    sample_idx = sort(sample_idx);


    [t_FR, qdot_FR] = compute_fay_riddell_heating( ...
        t(sample_idx), V(sample_idx), h(sample_idx), rn, species_model);


    [qdot_FR_peak, idx_FR_peak] = max(qdot_FR);


    % Sutton-Graves recomputed at the same points using ATM.atm density
    % for a consistent freestream comparison.
    qdot_SG_ATM = nan(size(t_FR));
    for k = 1:numel(t_FR)
        [~, ~, rho_ATM_k] = ATM.atm(max(h(sample_idx(k)),0)/1000);
        qdot_SG_ATM(k) = k_SG*sqrt(rho_ATM_k/rn)*V(sample_idx(k))^3/1e4;
    end


    FR_mask = t >= t_FR(1) & t <= t_FR(end);
    qdot_FR_full = nan(size(t));
    qdot_FR_full(FR_mask) = interp1(t_FR, qdot_FR, t(FR_mask), 'pchip');
    Q_FR = trapz(t(FR_mask), qdot_FR_full(FR_mask));
end


drag_accel  = rho_t.*V.^2./(2*beta_t);
aero_accel  = drag_accel.*sqrt(1 + LD_used^2);
load_factor = aero_accel/g0;
[load_peak, idx_load] = max(load_factor);
```

```
%% -------------------------------------------------------------------
%% MACH HISTORY / DROGUE-REGION MATCH (Stardust validation)
%% -------------------------------------------------------------------
if strcmp(aero_mode, 'equiflow')
    M_t = nan(size(t));
    for k = 1:numel(t)
        [~, Tinf_k, ~] = ATM.atm(max(h(k),0)/1000);
        a_k = sqrt(gamma_air*R_air*Tinf_k);
        M_t(k) = V(k)/a_k;
    end


    idx_Mref = find(M_t(1:end-1) >= reference.drogue_M & M_t(2:end) <
reference.drogue_M, 1, 'first');


    t_Mref = NaN; h_Mref = NaN; V_Mref = NaN;
    if ~isempty(idx_Mref)
        M_pair = M_t(idx_Mref:idx_Mref+1);
        t_pair = t(idx_Mref:idx_Mref+1);
        t_Mref = interp1(flip(M_pair), flip(t_pair), reference.drogue_M);
        h_Mref = interp1(t, h, t_Mref);
        V_Mref = interp1(t, V, t_Mref);
    end


    validation_mask = M_t >= reference.drogue_M;
end


%% -------------------------------------------------------------------
%% PLOTS
%% -------------------------------------------------------------------
figure;
subplot(3,2,1); plot(t, V/1000); xlabel('Time (s)'); ylabel('V (km/s)');
grid on;
subplot(3,2,2); plot(t, gamma*180/pi); xlabel('Time (s)'); ylabel('\gamma
(deg)'); grid on;
subplot(3,2,3); plot(t, h/1000); xlabel('Time (s)'); ylabel('h (km)');
grid on;
subplot(3,2,4); plot(t, load_factor); xlabel('Time (s)'); ylabel('Load
factor (g)'); grid on;
subplot(3,2,5); plot(t, qdot_s, t, qdot_rad, t, qdot_total); hold on;
if strcmp(aero_mode, 'equiflow')
    plot(t_FR, qdot_FR, 'o-', 'MarkerSize',5, 'LineWidth',1.2);
```

```
    legend('Convective (Sutton-Graves)', 'Radiative (Tauber-Sutton)',
'Total', 'Fay-Riddell (sampled)', 'Location','best');
else
    legend('Convective (Sutton-Graves)', 'Radiative (Tauber-Sutton)',
'Total', 'Location','best');
end
xlabel('Time (s)'); ylabel('qdot (W/cm^2)'); grid on;
subplot(3,2,6); plot(V/1000, h/1000); set(gca,'XDir','reverse');
xlabel('V (km/s)'); ylabel('h (km)'); grid on;


if strcmp(motion_mode, 'nonplanar')
    figure;
    downrange_km = Re*(theta-theta0)/1000;   % local flat-Earth
approximation
    crossrange_km = Re*(phi-phi0)/1000;
    subplot(2,1,1);
    plot(downrange_km, crossrange_km, 'LineWidth',1.5);
    xlabel('Downrange (km)'); ylabel('Crossrange (km)');
    title('Ground Track'); axis equal; grid on;
    subplot(2,1,2);
    plot(t, psi*180/pi, 'LineWidth',1.5);
    xlabel('Time (s)'); ylabel('\psi, heading (deg)');
    title('Heading vs. Time'); grid on;
end


if strcmp(aero_mode, 'equiflow')
    figure;
    tiledlayout(2,2,'TileSpacing','compact','Padding','compact');


    nexttile;
    plot(V(validation_mask)/1000, h(validation_mask)/1000,
'LineWidth',1.5, 'DisplayName','Model');
    hold on;
    if isfinite(V_Mref)
        plot(V_Mref/1000, h_Mref/1000, 'o', 'MarkerSize',7,
'LineWidth',1.2, 'DisplayName','Model at M=1.23');
    end
    set(gca,'XDir','reverse');
    xlabel('Planet-relative velocity (km/s)'); ylabel('Altitude (km)');
    title('Altitude-Velocity Validation'); legend('Location','best'); grid
on;


    nexttile;
```

```
    plot(t(validation_mask), load_factor(validation_mask),
'LineWidth',1.5, 'DisplayName','Model');
    hold on;
    yline(reference.peak_g_BET, '--', 'BET peak = 32.89 g',
'DisplayName','BET peak');
    plot(t(idx_load), load_peak, 'o', 'MarkerSize',7, 'LineWidth',1.2,
'DisplayName','Model peak');
    xlabel('Time from EI (s)'); ylabel('Aerodynamic deceleration (g)');
    title('Peak-Deceleration Validation'); legend('Location','best'); grid
on;


    nexttile;
    plot(t(validation_mask), M_t(validation_mask), 'LineWidth',1.5,
'DisplayName','Model Mach');
    hold on;
    yline(reference.drogue_M, '--', 'Reference M=1.23');
    if isfinite(t_Mref)
        plot(t_Mref, reference.drogue_M, 'o', 'MarkerSize',7,
'LineWidth',1.2, 'DisplayName','Model crossing');
    end
    xlabel('Time from EI (s)'); ylabel('Mach number');
    title('Drogue-Region Mach Comparison'); legend('Location','best');
grid on;


    nexttile;
    plot(t(validation_mask), qdot_s(validation_mask), 'LineWidth',1.5);
    xlabel('Time from EI (s)'); ylabel('Sutton-Graves heat rate
(W/cm^2)');
    title('Model Aerothermal Prediction'); grid on;
end


%% ------------------------------------------------------------------
%% SUMMARY
%% ------------------------------------------------------------------
fprintf('\n--- RUN SUMMARY ---\n');
fprintf('Outcome: %s\n', outcome);
fprintf('motion_mode = %s, aero_mode = %s, atm_model = %s, gravity_model =
%s\n', ...
    motion_mode, aero_mode, atm_model, gravity_model);
if strcmp(motion_mode, 'nonplanar')
    fprintf('  Final heading = %.2f deg, downrange = %.2f km, crossrange =
%.2f km\n', ...
        rad2deg(psi(end)), Re*(theta(end)-theta0)/1000, Re*(phi(end)-
phi0)/1000);
end
```

```
if strcmp(aero_mode, 'equiflow') && ~isnan(freeze_time)
    fprintf('  CD frozen at Mach %.2f: t=%.2f s, CD=%.4f, beta=%.2f
kg/m^2\n', ...
        M_transonic_cutoff, freeze_time, CD_freeze, beta_freeze);
end
fprintf('Peak load factor: %.2f g at t=%.1f s (h=%.1f km, V=%.1f m/s)\n',
load_peak, t(idx_load), h(idx_load)/1000, V(idx_load));
fprintf('Peak stagnation heat rate: %.2f W/cm^2 at t=%.1f s (h=%.1f km,
V=%.1f m/s)\n', qdot_peak, t(idx_heat), h(idx_heat)/1000, V(idx_heat));
fprintf('Total stagnation heat load: %.2f J/cm^2\n', Q_total);
fprintf('Peak radiative heat rate (Tauber-Sutton): %.2f W/cm^2 at t=%.1f s
(V=%.2f km/s)\n', ...
    qdot_rad_peak, t(idx_rad_peak), V_kms(idx_rad_peak));
fprintf('Radiative heat load: %.2f J/cm^2 (%.1f%% of convective heat
load)\n', ...
    Q_rad_total, 100*Q_rad_total/Q_total);
fprintf('Peak combined (convective + radiative) heat rate: %.2f W/cm^2\n',
max(qdot_total));
if strcmp(aero_mode, 'equiflow')
    fprintf('\n--- FAY-RIDDELL COMPARISON ---\n');
    fprintf('Fay-Riddell peak (%d sampled points): %.2f W/cm^2 at t=%.2f
s\n', ...
        numel(t_FR), qdot_FR_peak, t_FR(idx_FR_peak));
    fprintf('Sutton-Graves at same point (exponential atmosphere): %.2f
W/cm^2\n', ...
        qdot_s(sample_idx(idx_FR_peak)));
    fprintf('Sutton-Graves at same point (ATM.atm density): %.2f
W/cm^2\n', ...
        qdot_SG_ATM(idx_FR_peak));
    fprintf('Integrated Fay-Riddell heat load: %.2f J/cm^2\n', Q_FR);
end


%% ----------------------------------------------------------------
%% STARDUST VALIDATION
%% ----------------------------------------------------------------
if strcmp(aero_mode, 'equiflow')
    if isfinite(t_Mref)
        drogue_time_error = t_Mref - reference.drogue_time;
        drogue_h_error     = h_Mref - reference.drogue_h;
    else
        drogue_time_error = NaN;
        drogue_h_error     = NaN;
    end


    % Closed-form ballistic check (Allen-Eggers), independent of the
    % numerical integration.
```

```
    n_max_closed_form = V0^2*sin(gamma0)/(53.3*H);


    fprintf('\n--- STARDUST VALIDATION (Desai & Qualls, 2010) ---\n');
    fprintf('Closed-form (Allen-Eggers, ballistic) peak deceleration =
%.2f g\n', n_max_closed_form);
    if isfinite(t_Mref)
        fprintf('Drogue-region (M=%.2f): model dt=%+.2f s, dh=%+.3f km vs.
reference\n', ...
            reference.drogue_M, drogue_time_error, drogue_h_error/1000);
    else
        fprintf('Model did not cross M = %.2f.\n', reference.drogue_M);
    end


    validation_metric = {'Entry velocity'; 'Entry flight-path angle';
'Entry altitude'; ...
        'Peak deceleration'; 'Time at Mach 1.23'; 'Altitude at Mach
1.23'};
    model_value = [V0/1000; rad2deg(gamma0); h0/1000; load_peak; t_Mref;
h_Mref/1000];
    reference_value = [reference.V_EI/1000; -reference.gamma_EI_paper_deg;
reference.h_EI/1000; ...
        reference.peak_g_BET; reference.drogue_time;
reference.drogue_h/1000];
    units = {'km/s'; 'deg'; 'km'; 'g'; 's'; 'km'};


    absolute_error = model_value - reference_value;
    percent_error = 100*absolute_error./reference_value;
    percent_error(1:3) = 0;   % entry conditions are prescribed inputs,
not predictions


    validation_table = table(validation_metric, model_value,
reference_value, absolute_error, percent_error, units);
    disp(validation_table);
    writetable(validation_table, 'Stardust_validation_results.csv');
end


% J2 gravity model vs. known equatorial/polar surface
% gravity reference values.
phi_ref = [0, pi/2];
phi_label = {'Equatorial (phi=0)', 'Polar (phi=90 deg)'};
g_ref = [9.780, 9.832];
```

```
g_model = nan(size(phi_ref));
for i = 1:length(phi_ref)
    g_model(i) = g_j2(phi_ref(i), 0, mu_earth, J2_earth, omega_earth, Re,
K_ellip);
end


g_err = 100*(g_model - g_ref)./g_ref;


T_j2_check = table(phi_label', rad2deg(phi_ref)', g_model', g_ref',
g_err', ...
    'VariableNames',
{'Latitude_case','Latitude_deg','g_model_ms2','g_ref_ms2','pct_err'});
writetable(T_j2_check, 'appendix_j2_gravity_verification.csv');




%% =====================================================================
%% LOCAL FUNCTIONS
%% =====================================================================


function [value, isterminal, direction] = ground_skip_event(t, x,
skip_altitude, gamma_cutoff)
% gamma_cutoff: Inf disables the gamma-cutoff event (planar mode).
% Nonplanar mode passes a real cutoff (85 deg) to terminate before
% EOM_6DOF's 1/cos(gamma) singularity.
    h     = x(3);
    gamma = x(2);
    value      = [h; h - skip_altitude; gamma_cutoff - gamma];
    isterminal = [1; 1; 1];
    direction  = [-1; +1; -1];
end


function [value, isterminal, direction] = mach_ground_skip_event(t, x,
gamma_air, R_air, M_cutoff, skip_altitude, gamma_cutoff)
    V     = x(1);
    h     = x(3);
    gamma = x(2);
    [~, Tinf, ~] = ATM.atm(h/1000);
    a_sound = sqrt(gamma_air*R_air*Tinf);
    M = V/a_sound;


    value      = [h; M - M_cutoff; h - skip_altitude; gamma_cutoff -
gamma];
```

```
    isterminal = [1; 1; 1; 1];
    direction  = [-1; -1; +1; -1];
end


function CD = lookup_CD_checked(CD_interp, V_grid, h_grid, V, h)
    h_q = max(h, h_grid(1));
    if V < V_grid(1) || V > V_grid(end) || h_q > h_grid(end)
        error('CD table query out of bounds: V=%.1f m/s, h=%.2f km', V,
h/1000);
    end
    CD = CD_interp(V, h_q);
    if ~isscalar(CD) || ~isfinite(CD) || CD <= 0
        error('Invalid interpolated CD at V=%.1f m/s, h=%.3f km.', V,
h/1000);
    end
end


function rho2 = extract_shock_density(shock)
    candidate_names = {'Density','density','Rho','rho'};
    rho2 = NaN;
    for k = 1:numel(candidate_names)
        name = candidate_names{k};
        if isstruct(shock) && isfield(shock, name)
            rho2 = shock.(name);
            break;
        elseif isobject(shock) && isprop(shock, name)
            rho2 = shock.(name);
            break;
        end
    end
    if ~isscalar(rho2) || ~isfinite(rho2) || rho2 <= 0
        error('Could not obtain a valid post-shock density from
EquiFlow.');
    end
end


function CD_interp = build_stardust_CD_table(V_grid, h_grid, rn, rb,
delta_c, species_model, ...
    gamma_air, R_air, T2_react_threshold, M_transonic_cutoff, d_veh)
% Precomputes CD(V,h): Modified Newtonian (continuum) bridged to
% free-molecular via Knudsen number. See report Sec. 2.1., 2.4.


    ef = EquiFlow(species_model);
```

```
    nV = length(V_grid);
    nH = length(h_grid);
    CD_table = zeros(nV, nH);
    total_points = nV*nH;
    points_done = 0;


    area_ratio_term = (rn/rb)^2;


    kB   = 1.380649e-23;   % J/K
    d_N2 = 364e-12;        % m
    Tw   = 300;            % K


    fprintf('Building %dx%d CD(V,h) table (%d points)...\n', nV, nH,
total_points);
    build_tic = tic;
    next_progress_pct = 10;


    for j = 1:nH
        h_km = h_grid(j)/1000;
        [Pinf, Tinf, Rhoinf] = ATM.atm(h_km);
        a_sound  = sqrt(gamma_air*R_air*Tinf);
        V_cutoff = M_transonic_cutoff*a_sound;


        lambda = kB*Tinf / (pi*sqrt(2)*d_N2^2*Pinf);
        Kn = lambda/d_veh;


        for i = 1:nV
            V_eval = max(V_grid(i), V_cutoff);
            M = V_eval/a_sound;


            P2_P1_cpg     = 1 + (2*gamma_air/(gamma_air+1))*(M^2 - 1);
            rho2_rho1_cpg = ((gamma_air+1)*M^2) / ((gamma_air-1)*M^2 + 2);
            T2_cpg        = Tinf * (P2_P1_cpg / rho2_rho1_cpg);


            if T2_cpg > T2_react_threshold
                shock = ef.NormalShock(Tinf, Pinf, Rhoinf, V_eval);
                rho2  = extract_shock_density(shock);
                if rho2 <= Rhoinf
                    error('Normal-shock density did not increase at V=%.1f
m/s, h=%.1f km.', V_eval, h_km);
                end
```

```
                epsilon = Rhoinf/rho2;
            else
                epsilon = 1/rho2_rho1_cpg;
            end
            Cpmax = 2 - epsilon;


            CA_cont = Cpmax*( 0.5*(1-sin(delta_c)^4)*area_ratio_term ...
                        + sin(delta_c)^2*(1 -
area_ratio_term*cos(delta_c)^2) );
            CA_cont = max(CA_cont, eps);


            if Kn <= 1e-3
                CA = CA_cont;
            else
                s = V_eval/sqrt(2*R_air*Tinf);
                Tw_ratio_term = sqrt(pi*Tw/Tinf);


                CD_FM_sphere = 2 + (2/(3*s))*Tw_ratio_term;
                CD_FM_cone = sin(delta_c)^2 +
(1/s)*Tw_ratio_term*sin(delta_c) ...
                    + 2*cos(delta_c)^2;
                CA_FM = area_ratio_term*CD_FM_sphere + (1-
area_ratio_term)*CD_FM_cone;


                if Kn >= 10
                    CA = CA_FM;
                else
                    phi = pi*(3/8 + (1/8)*log10(Kn));
                    CA = CA_cont + (CA_FM - CA_cont)*sin(phi)^2;
                end
            end


            CD_table(i,j) = max(CA, eps);
            points_done = points_done + 1;


            pct_done = 100*points_done/total_points;
            if pct_done >= next_progress_pct
                elapsed = toc(build_tic);
                est_total = elapsed/points_done*total_points;
                fprintf('  %.0f%% done, elapsed %.1f min, est. total %.1f
min\n', pct_done, elapsed/60, est_total/60);
                next_progress_pct = next_progress_pct + 10;
```

```
            end
        end
    end
    fprintf('CD(V,h) table build complete in %.1f min.\n',
toc(build_tic)/60);


    CD_interp = griddedInterpolant({V_grid, h_grid}, CD_table, 'linear',
'none');
end


function [t_out, qdot_FR] = compute_fay_riddell_heating( ...
        t_sample, V_sample, h_sample, rn, species_model)

% Fay–Riddell stagnation-point heating: equilibrium normal-shock/edge
% states, fully catalytic wall, Le=1, cold wall Tw=300K, MN stagnation
% pressure. See report Sec. 2.3.


    Tw = 300;             % K
    ef = EquiFlow(species_model);


    n = numel(t_sample);
    qdot_FR = nan(n,1);


    for k = 1:n
        Vinf = V_sample(k);
        h_km = max(h_sample(k),0)/1000;


        [Pinf, Tinf, Rhoinf] = ATM.atm(h_km);


        shock = ef.NormalShock(Tinf, Pinf, Rhoinf, Vinf);


        rho2 = shock.Density;
        V2   = shock.DownstreamVelocity;


        if ~isfinite(rho2) || rho2 <= Rhoinf
            error(['Invalid post-shock density at t=%.3f s: ' ...
                   'rho2=%.6e, rhoinf=%.6e kg/m^3.'], ...
                   t_sample(k), rho2, Rhoinf);
        end

```

```
        if ~isfinite(V2) || V2 < 0
            error('Invalid downstream velocity at t=%.3f s.',
t_sample(k));
        end


        epsilon = Rhoinf/rho2;
        Cpmax   = 2 - epsilon;


        qinf = 0.5*Rhoinf*Vinf^2;
        Pe   = Pinf + Cpmax*qinf;


        h0e = shock.Enthalpy + 0.5*V2^2;


        edge = ef.SolveHP(h0e, Pe, Transport=true);


        rhoe = edge.Density;
        mue  = edge.Viscosity;
        Cpe  = edge.Cp;
        kfe  = edge.ThermalConductivity;


        if any(~isfinite([rhoe, mue, Cpe, kfe])) || ...
                rhoe <= 0 || mue <= 0 || Cpe <= 0 || kfe <= 0
            error('Invalid equilibrium edge properties at t=%.3f s.',
t_sample(k));
        end


        wall = ef.SolveTP(Tw, Pe, Transport=true);


        rhow = wall.Density;
        muw  = wall.Viscosity;
        hw   = wall.Enthalpy;


        if any(~isfinite([rhow, muw, hw])) || rhow <= 0 || muw <= 0
            error('Invalid equilibrium wall properties at t=%.3f s.',
t_sample(k));
        end


        Pr_e = mue*Cpe/kfe;
```

```matlab
        due_dx = (1/rn)*sqrt(2*(Pe-Pinf)/rhoe);
        enthalpy_difference = h0e - hw;


        if ~isfinite(Pr_e) || Pr_e <= 0
            error('Invalid edge Prandtl number at t=%.3f s.',
t_sample(k));
        end


        if ~isfinite(due_dx) || due_dx <= 0
            error('Invalid stagnation-point velocity gradient at t=%.3f
s.', t_sample(k));
        end


        if ~isfinite(enthalpy_difference) || enthalpy_difference <= 0
            error(['Nonpositive h0e-hw at t=%.3f s: ' ...
                   'h0e=%.6e, hw=%.6e J/kg.'], ...
                   t_sample(k), h0e, hw);
        end


        qdot_FR(k) = 0.763*Pr_e^(-0.6)*(rhoe*mue)^0.4*(rhow*muw)^0.1 ...
            *sqrt(due_dx)*enthalpy_difference/1e4;


        if ~isfinite(qdot_FR(k)) || qdot_FR(k) < 0
            error('Invalid Fay-Riddell heat flux at t=%.3f s.',
t_sample(k));
        end
    end


    t_out = t_sample(:);
end


function g = g_j2(phi, h, mu, J2, omega, r_equator, K)
% J2 oblateness gravity with rotation and ellipticity (Eq. 25-27).
    R0 = r_equator*(1 - K*sin(phi)^2);
    R  = R0 + h;
    D  = K*(1 - h/R0)*sin(2*phi);
    g = (mu/R^2)*(1 - 0.75*J2*(1 - 3*cos(2*phi))) -
R*omega^2*cos(phi)*cos(phi-D);
end


function rho = get_density(h, atm_model, rho0, H)
```

```
    switch atm_model
        case 'exponential'
            rho = rho0*exp(-h/H);
        case 'layered'
            [~, ~, rho] = std_atm_layered(h);
        otherwise
            error('Unknown atm_model: %s', atm_model);
    end
end


function fV = tauber_sutton_fV(V, planet)
% Tabulated Tauber-Sutton f(V), Table 1 (Tauber & Sutton, 1991), V in
% m/s. Log-space pchip interpolation.
    switch planet
        case 'earth'
            V_table = [9000,9250,9500,9750,10000,10250,10500,10750,11000,
...

11500,12000,12500,13000,13500,14000,14500,15000,15500,16000];
            fV_table =
[1.5,4.3,9.7,19.5,35,55,81,115,151,238,359,495,660,850,1065,1313,1550,1780
,2040];
        case 'mars'
            V_table =
[6000,6150,6300,6500,6700,6900,7000,7200,7400,7600,7800,8000,8200,8400,860
0,8800,9000];
            fV_table =
[0.2,1.0,1.95,3.42,5.1,7.1,8.1,10.2,12.5,14.8,17.1,19.2,21.4,24.1,26.0,28.
9,32.8];
        otherwise
            error('Unknown planet for tauber_sutton_fV: %s', planet);
    end


    fV = exp(interp1(V_table, log(fV_table), V, 'pchip', 'extrap'));
end


function [T, P, rho] = std_atm_layered(h_m)
% Layered US Standard Atmosphere: 1976 (0-86 km, geopotential, 7
% layers) + 1962 extension (86-150 km, geometric, layer-varying MW).
% See report Sec. 2.5.


    g0  = 9.80665;
    R   = 287.05;
    r_E = 6356766;      % m
```

```
    H_geop = h_m * r_E / (r_E + h_m);


    H_base = [0, 11000, 20000, 32000, 47000, 51000, 71000, 86000];
    T_base = [288.15, 216.65, 216.65, 228.65, 270.65, 270.65, 214.65,
186.87];
    L_1976 = [-6.5e-3, 0, 1e-3, 2.8e-3, 0, -2.8e-3, -2e-3, 0];


    P_base = zeros(size(H_base));
    P_base(1) = 101325;
    for i = 2:length(H_base)
        dHb = H_base(i) - H_base(i-1);
        if abs(L_1976(i-1)) < 1e-12
            P_base(i) = P_base(i-1) * exp(-g0*dHb/(R*T_base(i-1)));
        else
            P_base(i) = P_base(i-1) * (T_base(i)/T_base(i-1))^(-
g0/(R*L_1976(i-1)));
        end
    end


    if h_m < 86000
        layer = find(H_geop >= H_base, 1, 'last');
        T_l = T_base(layer);
        L_l = L_1976(layer);
        dH = H_geop - H_base(layer);
        T = T_l + L_l*dH;
        if abs(L_l) < 1e-12
            P = P_base(layer) * exp(-g0*dH/(R*T_l));
        else
            P = P_base(layer) * (T/T_l)^(-g0/(R*L_l));
        end
        rho = P/(R*T);
        return;
    end


    Z_base   = [86, 100, 110, 120, 150] * 1e3;
    Tm_base  = [186.945, 210.65, 260.65, 360.65];
    Lz       = [1.6481e-3, 5.0e-3, 10.0e-3, 20.0e-3];
    P_base2  = [0.34313, 0.030075, 0.0073544, 0.0025217];
    MW       = [28.9644, 28.88, 28.56, 28.08];


    R_bar = 8314.0;
```

```
    b     = 3.31e-7;


    Z = h_m;
    if Z >= Z_base(end)
        layer2 = 4;
    else
        layer2 = find(Z >= Z_base(1:4), 1, 'last');
    end


    Tm_i = Tm_base(layer2);
    Lz_i = Lz(layer2);
    P_i  = P_base2(layer2);
    MW_i = MW(layer2);
    z_i  = Z_base(layer2);
    R_i  = R_bar / MW_i;


    T = Tm_i + Lz_i*(Z - z_i);
    rho_i = P_i * MW_i / (R_bar * Tm_i);


    POW_p   = -(g0/(R_i*Lz_i)) * (1 + b*(Tm_i/Lz_i - z_i));
    POW_rho = -(g0/(R_i*Lz_i)) * (R_i*Lz_i/g0 + 1 + b*(Tm_i/Lz_i - z_i));
    exp_term = exp(g0*b/(R_i*Lz_i) * (Z - z_i));


    P   = P_i   * (T/Tm_i)^POW_p   * exp_term;
    rho = rho_i * (T/Tm_i)^POW_rho * exp_term;
end


function xdot = EOM_3DOF(t, x, params)
% 3-DOF planar EOM (Eq. 1-3). gamma positive below horizontal.
    V     = x(1);
    gamma = x(2);
    h     = x(3);


    if V <= 0
        error('Nonpositive velocity at t=%.6f s.', t);
    end


    h_env = max(h, 0);
    Re    = params.Re;
    beta  = params.beta(V, h_env);
    LD    = params.LD(V, h_env);
```

```
    sigma = params.sigma(t);
    g     = params.g_fun(h_env);
    rho   = params.rho_fun(h_env);


    dVdt     = -rho*V^2/(2*beta) + g*sin(gamma);
    dgammadt = -V*cos(gamma)/(Re + h) - (rho*V/(2*beta))*LD*cos(sigma) +
g*cos(gamma)/V;
    dhdt     = -V*sin(gamma);


    xdot = [dVdt; dgammadt; dhdt];
end


function xdot = EOM_6DOF(t, x, params)
% 6-state non-planar EOM, Earth-fixed rotating frame (Eq. 8-13).
% gamma positive below horizontal (course convention negated; see
% report Sec. 2.2).
    V     = x(1);
    gamma = x(2);
    h     = x(3);
    psi   = x(4);
    phi   = x(5);


    if V <= 0
        error('Nonpositive velocity at t=%.6f s.', t);
    end


    h_env = max(h, 0);
    Re    = params.Re;
    beta  = params.beta(V, h_env);
    LD    = params.LD(V, h_env);
    sigma = params.sigma(t);
    if isfield(params,'gravity_model') && strcmp(params.gravity_model,
'j2')
        g = params.g_j2_fun(phi, h_env);
    else
        g = params.g_fun(h_env);
    end
    rho   = params.rho_fun(h_env);
    r     = Re + h;
    omega = params.omega;


    dVdt = -rho*V^2/(2*beta) + g*sin(gamma) ...
```

```
        - omega^2*r*cos(phi)*(sin(gamma)*cos(phi) +
cos(gamma)*sin(phi)*sin(psi));


    dgammadt = -(rho*V/(2*beta))*LD*cos(sigma) + g*cos(gamma)/V -
V*cos(gamma)/r ...
        - 2*omega*cos(phi)*cos(psi) ...
        - (omega^2*r/V)*cos(phi)*(cos(gamma)*cos(phi) -
sin(gamma)*sin(phi)*sin(psi));


    dhdt = -V*sin(gamma);


    dpsidt = (rho*V/(2*beta))*LD*sin(sigma)/cos(gamma) ...
        - V*cos(gamma)*cos(psi)*tan(phi)/r ...
        - 2*omega*(tan(gamma)*cos(phi)*sin(psi) + sin(phi)) ...
        - (omega^2*r/(V*cos(gamma)))*sin(phi)*cos(phi)*cos(psi);


    dphidt   = (V/r)*cos(gamma)*sin(psi);
    dthetadt = (V/r)*cos(gamma)*cos(psi)/cos(phi);


    xdot = [dVdt; dgammadt; dhdt; dpsidt; dphidt; dthetadt];
end
```

F. OAT Parameter Sweep MATLAB Script

```
%% AE 6355 - Individual Project - Parameter Sensitivity Study
% One-at-a-time (OAT) sweep of entry flight path angle, ballistic
% coefficient, L/D, bank angle, and entry velocity, holding all other
% parameters at a nominal baseline.
% Runs in constant-beta/L-D mode (not the EquiFlow-based aero used in base
% project), consistent with treating these as direct input parameters for
% a generic sensitivity study independent of the Stardust-specific
validation case. See Appendix Section C.

clear; clc; close all;


%% -----------------------------------------------------------------
%% BASELINE (NOMINAL) CASE AND OAT RANGES
%% -----------------------------------------------------------------
% Baseline uses a nonzero lifting condition (not the ballistic Stardust
% case) so both the L/D and bank-angle sweeps produce meaningful
% trajectory changes; a ballistic (L/D=0) baseline would make sigma
% have no physical effect at all.
```

```
baseline.gamma_e_deg = 8.21;       % deg, positive below local horizontal
baseline.beta        = 61;          % kg/m^2
baseline.LD          = 0.30;
baseline.sigma_deg   = 45;          % deg
baseline.Ve          = 12799;       % m/s


% OAT min/max ranges (assumptions, documented in the report). L/D and
% bank-angle ranges are centered on the baseline so both perturbation
% directions are informative.
ranges.gamma_e_deg = [1, 15];       % deg; 1 deg intended to demonstrate
skip-out
ranges.beta        = [20, 200];    % kg/m^2
ranges.LD          = [0, 0.60];    % centered on baseline (0.30)
ranges.sigma_deg   = [0, 90];      % deg; centered on baseline (45)
ranges.Ve          = [7500, 13000];% m/s


% Build the 11-case OAT matrix: 1 baseline + (5 parameters x 2 endpoints).
case_names = {};
case_params = {};


case_names{end+1} = 'baseline';
case_params{end+1} = baseline;


param_fields = {'gamma_e_deg','beta','LD','sigma_deg','Ve'};
for k = 1:numel(param_fields)
    fname = param_fields{k};
    for endpoint = 1:2
        c = baseline;
        c.(fname) = ranges.(fname)(endpoint);
        label = sprintf('%s_%s', fname,
ternary(endpoint==1,'low','high'));
        case_names{end+1} = label;
        case_params{end+1} = c;
    end
end

%% ------------------------------------------------------------------
%% RUN ALL CASES
%% ------------------------------------------------------------------
n_cases = numel(case_names);
results = struct([]);


for i = 1:n_cases
```

```
    c = case_params{i};
    r = run_entry_trajectory(c.Ve, c.gamma_e_deg, c.beta, c.LD,
c.sigma_deg);
    r.case_name = case_names{i};
    r.gamma_e_deg = c.gamma_e_deg;
    r.beta = c.beta;
    r.LD = c.LD;
    r.sigma_deg = c.sigma_deg;
    r.Ve = c.Ve;
    results = [results, r];
    fprintf('[%2d/%2d] %-16s -> %s (peak load %.1f g, peak qdot %.1f
W/cm^2, h_min=%.1f km)\n', ...
        i, n_cases, case_names{i}, r.outcome, r.peak_load, r.peak_qdot,
r.h_min/1000);
end


%% ------------------------------------------------------------------
%% SUMMARY TABLE
%% ------------------------------------------------------------------
fprintf('\n=== OAT SENSITIVITY STUDY SUMMARY ===\n');
fprintf('%-16s %8s %8s %6s %8s %9s | %-20s %9s %10s %9s %9s\n', ...
    'case', 'gamma_e', 'beta', 'LD', 'sigma', 'Ve', 'outcome', 'peak_g',
'peak_qdot', 'Q_total', 'h_min_km');
for i = 1:n_cases
    r = results(i);
    fprintf('%-16s %8.2f %8.1f %6.2f %8.1f %9.0f | %-20s %9.2f %10.2f
%9.1f %9.2f\n', ...
        r.case_name, r.gamma_e_deg, r.beta, r.LD, r.sigma_deg, r.Ve, ...
        r.outcome, r.peak_load, r.peak_qdot, r.Q_total, r.h_min/1000);
end


%% ------------------------------------------------------------------
%% CSV EXPORT
%% ------------------------------------------------------------------
case_name   = {results.case_name}';
gamma_e_deg = [results.gamma_e_deg]';
beta        = [results.beta]';
LD          = [results.LD]';
sigma_deg   = [results.sigma_deg]';
Ve          = [results.Ve]';
outcome     = {results.outcome}';
peak_load   = [results.peak_load]';
peak_load_t = [results.peak_load_t]';
peak_load_h = [results.peak_load_h]';
peak_load_V = [results.peak_load_V]';
peak_qdot   = [results.peak_qdot]';
```

```matlab
peak_qdot_t = [results.peak_qdot_t]';
peak_qdot_h = [results.peak_qdot_h]';
peak_qdot_V = [results.peak_qdot_V]';
Q_total     = [results.Q_total]';
h_min       = [results.h_min]';
t_h_min     = [results.t_h_min]';
V_h_min     = [results.V_h_min]';
t_final     = [results.t_final]';
h_final     = [results.h_final]';
V_final     = [results.V_final]';


results_table = table(case_name, gamma_e_deg, beta, LD, sigma_deg, Ve,
outcome, ...
    peak_load, peak_load_t, peak_load_h, peak_load_V, ...
    peak_qdot, peak_qdot_t, peak_qdot_h, peak_qdot_V, Q_total, ...
    h_min, t_h_min, V_h_min, t_final, h_final, V_final);
writetable(results_table, 'OAT_sweep_results.csv');
fprintf('\nResults written to OAT_sweep_results.csv\n');


%% -----------------------------------------------------------------
%% LOCAL FUNCTIONS
%% -----------------------------------------------------------------


function out = ternary(cond, a, b)
    if cond, out = a; else, out = b; end
end


function result = run_entry_trajectory(V0, gamma0_deg, beta_val, LD_val,
sigma_deg)
% Runs one constant-beta/L-D entry trajectory and returns summary
% metrics. Shares the EOM, exponential atmosphere, inverse-square
% gravity, and Sutton-Graves heating conventions of EDL_BaseProject.m.


    Re  = 6378150;      % m
    g0  = 9.81;          % m/s^2
    rho0 = 1.225;        % kg/m^3
    H   = 7200;          % m
    h0  = 120000;        % m, entry interface altitude


    % Skip-out threshold sits above h0 so the upward-crossing event is
    % armed correctly; kept tight (h0+100) so genuine skip-out is
    % detected shortly after re-crossing the entry interface.
```

```
    skip_altitude = h0 + 100;   % m
    k_SG = 1.7415e-4;
    rn  = 0.2202;        % m, Stardust nose radius (reused for heating
calc)


    gamma0 = gamma0_deg*pi/180;
    sigma0 = sigma_deg*pi/180;


    params.Re      = Re;
    params.beta    = @(V,h) beta_val;
    params.LD      = @(V,h) LD_val;
    params.sigma   = @(t) sigma0;
    params.g_fun   = @(h) g0*(Re/(Re+max(h,0)))^2;
    params.rho_fun = @(h) rho0*exp(-max(h,0)/H);


    x0 = [V0; gamma0; h0];
    tspan = [0 8000];   % extended ceiling for shallow/grazing entries
    opts = odeset('RelTol',1e-8, 'AbsTol',[1e-3 1e-10 1e-3],
'MaxStep',0.5, ...
        'Events', @(t,x) ground_skip_event(t,x,skip_altitude));
    [t, x, te, xe, ie] = ode45(@(t,x) EOM_3DOF(t, x, params), tspan, x0,
opts);


    if isempty(ie)
        outcome = 'time limit';
        te = t(end);
        xe = x(end,:);
    elseif ie(end) == 1
        outcome = 'ground impact';
    else
        outcome = 'atmospheric skip-out';
    end


    V = x(:,1); h = x(:,3);


    rho_t = rho0*exp(-h/H);
    qdot_s = k_SG*sqrt(rho_t/rn).*V.^3 / 1e4;   % W/cm^2
    Q_total = trapz(t, qdot_s);                  % J/cm^2
    [qdot_peak, idx_heat] = max(qdot_s);


    drag_accel = rho_t.*V.^2./(2*beta_val);
    aero_accel = drag_accel*sqrt(1+LD_val^2);
```

```
    load_factor = aero_accel/g0;
    [load_peak, idx_load] = max(load_factor);
    [h_min, idx_min_h] = min(h);


    result.outcome     = outcome;
    result.t_final     = te;
    result.V_final     = xe(1);
    result.gamma_final = xe(2);
    result.h_final      = xe(3);
    result.peak_load    = load_peak;
    result.peak_load_t  = t(idx_load);
    result.peak_load_h  = h(idx_load);
    result.peak_load_V  = V(idx_load);
    result.peak_qdot    = qdot_peak;
    result.peak_qdot_t  = t(idx_heat);
    result.peak_qdot_h  = h(idx_heat);
    result.peak_qdot_V  = V(idx_heat);
    result.Q_total      = Q_total;
    result.h_min        = h_min;
    result.t_h_min      = t(idx_min_h);
    result.V_h_min      = V(idx_min_h);
end


function [value, isterminal, direction] = ground_skip_event(t, x,
skip_altitude)
% ie: 1 = ground impact (h=0, decreasing), 2 = atmospheric skip-out
% (h crosses skip_altitude, increasing).
    h = x(3);
    value      = [h; h - skip_altitude];
    isterminal = [1; 1];
    direction  = [-1; +1];
end


function xdot = EOM_3DOF(t, x, params)
% 3-DOF planar EOM with bank angle. gamma positive below horizontal.
    V     = x(1);
    gamma = x(2);
    h     = x(3);


    if V <= 0
        error('Nonpositive velocity at t=%.6f s.', t);
    end
```

```
    h_env = max(h, 0);
    Re    = params.Re;
    beta  = params.beta(V, h_env);
    LD    = params.LD(V, h_env);
    sigma = params.sigma(t);
    g     = params.g_fun(h_env);
    rho   = params.rho_fun(h_env);


    dVdt     = -rho*V^2/(2*beta) + g*sin(gamma);
    dgammadt = -V*cos(gamma)/(Re + h) - (rho*V/(2*beta))*LD*cos(sigma) +
g*cos(gamma)/V;
    dhdt     = -V*sin(gamma);


    xdot = [dVdt; dgammadt; dhdt];
end
```

G. Monte Carlo Simulation Script

```
%% AE 6355 - Monte Carlo Landing Ellipse
% Minor Upgrade #4: disperses atmospheric density, entry flight path
% angle, entry velocity, and initial latitude/longitude to produce a
% landing footprint (ellipse) for the Stardust-conditions case, using
% the same EquiFlow-based aerodynamics and non-planar EOM as the main
% validation run. See report Sec. 2.8, 3.8.
%
% Dispersions (1-sigma, Gaussian, truncated at +/-3 sigma):
%   Atmospheric density bias : 10%   (single multiplicative bias per
%                                      trial, held constant along the
%                                      trajectory)
%   Entry flight path angle  : 0.2 deg
%   Entry velocity           : 10 m/s
%   Initial latitude         : 0.05 deg
%   Initial longitude        : 0.05 deg


clear; clc; close all;
rng(1);   % fixed seed for reproducibility


%% ------------------------------------------------------------------
%% NOMINAL CASE (matches Stardust configuration)
%% ------------------------------------------------------------------
g0 = 9.81;
Re = 6378150;
omega_earth = 7.292115147e-5;   % rad/s
```

```
gamma_cutoff = deg2rad(85);     % avoids EOM_6DOF's 1/cos(gamma)
singularity; see report Sec. 2.2


V0_nom     = 12799;
gamma0_nom = deg2rad(8.21);
h0         = 6503.14e3 - Re;
psi0_nom   = deg2rad(102.9);
phi0_nom   = 0;
theta0_nom = 0;


m_veh   = 46;
d_veh   = 0.8128;
rn      = 0.2202;
rb      = d_veh/2;
delta_c = 60*pi/180;
A_ref   = pi*rb^2;


gamma_air           = 1.4;
R_air               = 287;
T2_react_threshold  = 800;
M_transonic_cutoff  = 1.2;


skip_altitude = h0 + 100;


V_grid = linspace(0, 14000, 20);
h_grid = linspace(0, 130000, 20);
assert(skip_altitude <= max(h_grid), 'skip_altitude must lie inside the
aerodynamic-table altitude range.');


cpmax_formula_version = 3;
species_model = 'earth10';
CD_table_cache_file = 'stardust_CD_table_cache.mat';


cache_valid = false;
if isfile(CD_table_cache_file)
    try
        loaded = load(CD_table_cache_file, 'CD_interp', 'V_grid',
'h_grid', 'rn', 'rb', 'delta_c', ...
            'gamma_air', 'R_air', 'T2_react_threshold',
'M_transonic_cutoff', ...
            'cpmax_formula_version', 'species_model');
```

```
        cache_valid = isequal(loaded.V_grid, V_grid) &&
isequal(loaded.h_grid, h_grid) ...
            && loaded.rn==rn && loaded.rb==rb && loaded.delta_c==delta_c
...
            && loaded.gamma_air==gamma_air && loaded.R_air==R_air ...
            && loaded.T2_react_threshold==T2_react_threshold ...
            && loaded.M_transonic_cutoff==M_transonic_cutoff ...
            && loaded.cpmax_formula_version==cpmax_formula_version ...
            && strcmp(loaded.species_model, species_model);
    catch
        cache_valid = false;
    end
end
if cache_valid
    CD_interp = loaded.CD_interp;
    fprintf('Loaded cached CD(V,h) table from %s.\n',
CD_table_cache_file);
else
    fprintf('Building CD(V,h) table...\n');
    CD_interp = build_stardust_CD_table(V_grid, h_grid, rn, rb, delta_c,
species_model, ...
        gamma_air, R_air, T2_react_threshold, M_transonic_cutoff, d_veh);
    save(CD_table_cache_file, 'CD_interp', 'V_grid', 'h_grid', 'rn', 'rb',
'delta_c', ...
        'gamma_air', 'R_air', 'T2_react_threshold', 'M_transonic_cutoff',
...
        'cpmax_formula_version', 'species_model', 'd_veh');
end


rho0 = 1.225;
H    = 7200;


%% -----------------------------------------------------------------
%% DISPERSIONS (1-sigma)
%% -----------------------------------------------------------------
n_trials = 2000;


sigma_rho_frac = 0.10;          % fractional density bias
sigma_gamma    = deg2rad(0.2); % rad
sigma_V        = 10;            % m/s
sigma_phi      = deg2rad(0.05);% rad
sigma_theta    = deg2rad(0.05);% rad


randn_clipped = @(sigma) max(min(sigma*randn(), 3*sigma), -3*sigma);
```

```matlab
%% ------------------------------------------------------------------
%% MONTE CARLO LOOP
%% ------------------------------------------------------------------
ode_settings = {'RelTol',1e-8, 'AbsTol',[1e-3 1e-10 1e-3 1e-10 1e-10 1e-
10], 'MaxStep',0.5};
tspan = [0 2000];


phi_land   = nan(n_trials,1);
theta_land = nan(n_trials,1);
outcome_list = cell(n_trials,1);


fprintf('Running %d Monte Carlo trials...\n', n_trials);
mc_tic = tic;
for i = 1:n_trials
    delta_rho = randn_clipped(sigma_rho_frac);
    gamma0_i  = gamma0_nom + randn_clipped(sigma_gamma);
    V0_i      = V0_nom + randn_clipped(sigma_V);
    phi0_i    = phi0_nom + randn_clipped(sigma_phi);
    theta0_i  = theta0_nom + randn_clipped(sigma_theta);


    params.Re = Re;
    params.beta = @(V,h) m_veh./(lookup_CD_checked(CD_interp, V_grid,
h_grid, V, h).*A_ref);
    params.LD   = @(V,h) 0;
    params.sigma = @(t) 0;
    params.g_fun = @(h) g0*(Re/(Re+max(h,0)))^2;
    params.rho_fun = @(h) rho0*(1+delta_rho)*exp(-max(h,0)/H);
    params.gravity_model = 'inverse_square';
    params.omega = omega_earth;


    x0_i = [V0_i; gamma0_i; h0; psi0_nom; phi0_i; theta0_i];


    [outcome, phi_f, theta_f] = run_dispersed_trajectory(x0_i, params,
tspan, ode_settings, ...
        skip_altitude, gamma_air, R_air, M_transonic_cutoff, CD_interp,
V_grid, h_grid, m_veh, A_ref, gamma_cutoff);


    outcome_list{i} = outcome;
    if strcmp(outcome, 'ground impact')
        phi_land(i) = phi_f;
        theta_land(i) = theta_f;
```

```
    end


    if mod(i,50)==0
        fprintf('  %d/%d trials complete (%.1f s elapsed)\n', i, n_trials,
toc(mc_tic));
    end
end
fprintf('Monte Carlo complete in %.1f min.\n', toc(mc_tic)/60);


n_impact = nnz(~isnan(phi_land));
n_skip   = nnz(strcmp(outcome_list, 'atmospheric skip-out'));
n_other  = n_trials - n_impact - n_skip;
fprintf('Outcomes: %d ground impact, %d atmospheric skip-out, %d other\n',
n_impact, n_skip, n_other);


%% -----------------------------------------------------------------
%% LANDING ELLIPSE (downrange/crossrange, local flat-Earth approx)
%% -----------------------------------------------------------------
valid = ~isnan(phi_land);
downrange_km  = Re*(theta_land(valid)-theta0_nom)/1000;
crossrange_km = Re*(phi_land(valid)-phi0_nom)/1000;


mean_dr = mean(downrange_km);
mean_cr = mean(crossrange_km);
cov_mat = cov([downrange_km, crossrange_km]);


fprintf('\n--- LANDING ELLIPSE SUMMARY (%d successful trials) ---\n',
n_impact);
fprintf('Mean landing point: downrange=%.2f km, crossrange=%.2f km\n',
mean_dr, mean_cr);
fprintf('Std dev: downrange=%.2f km, crossrange=%.2f km\n',
std(downrange_km), std(crossrange_km));


[eigvec, eigval] = eig(cov_mat);
[eigval_sorted, idx] = sort(diag(eigval), 'descend');
eigvec_sorted = eigvec(:, idx);
semi_major_1sigma = sqrt(eigval_sorted(1));
semi_minor_1sigma = sqrt(eigval_sorted(2));
ellipse_angle = atan2(eigvec_sorted(2,1), eigvec_sorted(1,1));


fprintf('1-sigma ellipse: semi-major=%.2f km, semi-minor=%.2f km,
orientation=%.1f deg from downrange axis\n', ...
```

```
    semi_major_1sigma, semi_minor_1sigma, rad2deg(ellipse_angle));


% Ellipse boundary curves (1, 2, 3-sigma), as x,y point sets for plotting
theta_circ = linspace(0, 2*pi, 100)';
R_rot = [cos(ellipse_angle) -sin(ellipse_angle); sin(ellipse_angle)
cos(ellipse_angle)];


ellipse_1s = R_rot * [semi_major_1sigma*1*cos(theta_circ)';
semi_minor_1sigma*1*sin(theta_circ)'];
ellipse_2s = R_rot * [semi_major_1sigma*2*cos(theta_circ)';
semi_minor_1sigma*2*sin(theta_circ)'];
ellipse_3s = R_rot * [semi_major_1sigma*3*cos(theta_circ)';
semi_minor_1sigma*3*sin(theta_circ)'];


T_mc_ellipses = table( ...
    mean_dr+ellipse_1s(1,:)', mean_cr+ellipse_1s(2,:)', ...
    mean_dr+ellipse_2s(1,:)', mean_cr+ellipse_2s(2,:)', ...
    mean_dr+ellipse_3s(1,:)', mean_cr+ellipse_3s(2,:)', ...
    'VariableNames',
{'x_1sigma','y_1sigma','x_2sigma','y_2sigma','x_3sigma','y_3sigma'});
writetable(T_mc_ellipses, 'stardust_mc_ellipses.csv');


% Summary statistics (one row)
T_mc_summary = table(mean_dr, mean_cr, std(downrange_km),
std(crossrange_km), ...
    semi_major_1sigma, semi_minor_1sigma, rad2deg(ellipse_angle),
n_impact, n_trials, ...
    'VariableNames',
{'mean_downrange_km','mean_crossrange_km','std_downrange_km', ...

'std_crossrange_km','semi_major_km','semi_minor_km','orientation_deg','n_i
mpact','n_trials'});
writetable(T_mc_summary, 'stardust_mc_summary.csv');


figure;
scatter(downrange_km, crossrange_km, 15, 'filled', 'MarkerFaceAlpha',
0.4); hold on;
theta_circ = linspace(0, 2*pi, 100);
for n_sigma = 1:3
    ellipse_pts = [semi_major_1sigma*n_sigma*cos(theta_circ);
semi_minor_1sigma*n_sigma*sin(theta_circ)];
    R_rot = [cos(ellipse_angle) -sin(ellipse_angle); sin(ellipse_angle)
cos(ellipse_angle)];
```

```
    ellipse_rot = R_rot*ellipse_pts;
    plot(mean_dr+ellipse_rot(1,:), mean_cr+ellipse_rot(2,:), 'LineWidth',
1.5, ...
        'DisplayName', sprintf('%d-sigma', n_sigma));
end
plot(mean_dr, mean_cr, 'r+', 'MarkerSize', 12, 'LineWidth', 2,
'DisplayName', 'Mean landing point');
xlabel('Downrange (km)'); ylabel('Crossrange (km)');
title(sprintf('Landing Ellipse (%d trials, %d successful)', n_trials,
n_impact));
legend('Location','best'); axis equal; grid on;




%% ======================================================================
%% LOCAL FUNCTIONS
%% ======================================================================


function [outcome, phi_f, theta_f] = run_dispersed_trajectory(x0, params,
tspan, ode_settings, ...
    skip_altitude, gamma_air, R_air, M_transonic_cutoff, CD_interp,
V_grid, h_grid, m_veh, A_ref, gamma_cutoff)
% Runs one two-segment (Mach-freeze) non-planar trajectory to impact,
% skip-out, or gamma cutoff. Returns only the outcome and final
latitude/longitude.
% gamma_cutoff is treated as a landing outcome, not a separate case,
% since the vehicle is falling nearly vertically by that point.


    opts1 = odeset(ode_settings{:}, 'Events', ...
        @(t,x)
mach_ground_skip_event(t,x,gamma_air,R_air,M_transonic_cutoff,skip_altitud
e,gamma_cutoff));
    [t1, x1, te1, xe1, ie1] = ode45(@(t,x) EOM_6DOF(t, x, params), tspan,
x0, opts1);


    if isempty(ie1)
        outcome = 'time limit'; xe = x1(end,:);
    elseif ie1(end) == 1
        outcome = 'ground impact'; xe = xe1;
    elseif ie1(end) == 3
        outcome = 'atmospheric skip-out'; xe = xe1;
    elseif ie1(end) == 4
        outcome = 'ground impact'; xe = xe1;   % gamma cutoff treated as
impact
```

```
    else
        V_freeze = xe1(1);
        h_freeze = xe1(3);
        CD_freeze = lookup_CD_checked(CD_interp, V_grid, h_grid, V_freeze,
h_freeze);
        beta_freeze = m_veh/(CD_freeze*A_ref);


        params2 = params;
        params2.beta = @(V,h) beta_freeze;


        opts2 = odeset(ode_settings{:}, 'Events', @(t,x)
ground_skip_event(t,x,skip_altitude,gamma_cutoff));
        [t2, x2, te2, xe2, ie2] = ode45(@(t,x) EOM_6DOF(t, x, params2),
[te1 tspan(2)], xe1, opts2);


        if isempty(ie2)
            outcome = 'time limit'; xe = x2(end,:);
        elseif ie2(end) == 1
            outcome = 'ground impact'; xe = xe2;
        elseif ie2(end) == 3
            outcome = 'ground impact'; xe = xe2;   % gamma cutoff treated
as impact
        else
            outcome = 'atmospheric skip-out'; xe = xe2;
        end
    end


    phi_f   = xe(5);
    theta_f = xe(6);
end


function [value, isterminal, direction] = ground_skip_event(t, x,
skip_altitude, gamma_cutoff)
    h     = x(3);
    gamma = x(2);
    value      = [h; h - skip_altitude; gamma_cutoff - gamma];
    isterminal = [1; 1; 1];
    direction  = [-1; +1; -1];
end


function [value, isterminal, direction] = mach_ground_skip_event(t, x,
gamma_air, R_air, M_cutoff, skip_altitude, gamma_cutoff)
```

```
    V     = x(1);
    h     = x(3);
    gamma = x(2);
    [~, Tinf, ~] = ATM.atm(h/1000);
    a_sound = sqrt(gamma_air*R_air*Tinf);
    M = V/a_sound;


    value      = [h; M - M_cutoff; h - skip_altitude; gamma_cutoff -
gamma];
    isterminal = [1; 1; 1; 1];
    direction  = [-1; -1; +1; -1];
end


function CD = lookup_CD_checked(CD_interp, V_grid, h_grid, V, h)
    h_q = max(h, h_grid(1));
    if V < V_grid(1) || V > V_grid(end) || h_q > h_grid(end)
        error('CD table query out of bounds: V=%.1f m/s, h=%.2f km', V,
h/1000);
    end
    CD = CD_interp(V, h_q);
    if ~isscalar(CD) || ~isfinite(CD) || CD <= 0
        error('Invalid interpolated CD at V=%.1f m/s, h=%.3f km.', V,
h/1000);
    end
end


function rho2 = extract_shock_density(shock)
    candidate_names = {'Density','density','Rho','rho'};
    rho2 = NaN;
    for k = 1:numel(candidate_names)
        name = candidate_names{k};
        if isstruct(shock) && isfield(shock, name)
            rho2 = shock.(name);
            break;
        elseif isobject(shock) && isprop(shock, name)
            rho2 = shock.(name);
            break;
        end
    end
    if ~isscalar(rho2) || ~isfinite(rho2) || rho2 <= 0
        error('Could not obtain a valid post-shock density from
EquiFlow.');
    end
end
```

```matlab
function CD_interp = build_stardust_CD_table(V_grid, h_grid, rn, rb,
delta_c, species_model, ...
    gamma_air, R_air, T2_react_threshold, M_transonic_cutoff, d_veh)
% Continuum (Modified Newtonian) bridged to free-molecular via Knudsen
% number. See report Sec. 2.1, 2.4.


    ef = EquiFlow(species_model);


    nV = length(V_grid);
    nH = length(h_grid);
    CD_table = zeros(nV, nH);
    total_points = nV*nH;
    points_done = 0;


    area_ratio_term = (rn/rb)^2;


    kB   = 1.380649e-23;   % J/K
    d_N2 = 364e-12;        % m
    Tw   = 300;            % K


    fprintf('Building %dx%d CD(V,h) table (%d points)...\n', nV, nH,
total_points);
    build_tic = tic;
    next_progress_pct = 10;


    for j = 1:nH
        h_km = h_grid(j)/1000;
        [Pinf, Tinf, Rhoinf] = ATM.atm(h_km);
        a_sound  = sqrt(gamma_air*R_air*Tinf);
        V_cutoff = M_transonic_cutoff*a_sound;


        lambda = kB*Tinf / (pi*sqrt(2)*d_N2^2*Pinf);
        Kn = lambda/d_veh;


        for i = 1:nV
            V_eval = max(V_grid(i), V_cutoff);
            M = V_eval/a_sound;


            P2_P1_cpg     = 1 + (2*gamma_air/(gamma_air+1))*(M^2 - 1);
            rho2_rho1_cpg = ((gamma_air+1)*M^2) / ((gamma_air-1)*M^2 + 2);
```

```
            T2_cpg        = Tinf * (P2_P1_cpg / rho2_rho1_cpg);


            if T2_cpg > T2_react_threshold
                shock = ef.NormalShock(Tinf, Pinf, Rhoinf, V_eval);
                rho2  = extract_shock_density(shock);
                if rho2 <= Rhoinf
                    error('Normal-shock density did not increase at V=%.1f
m/s, h=%.1f km.', V_eval, h_km);
                end
                epsilon = Rhoinf/rho2;
            else
                epsilon = 1/rho2_rho1_cpg;
            end
            Cpmax = 2 - epsilon;


            CA_cont = Cpmax*( 0.5*(1-sin(delta_c)^4)*area_ratio_term ...
                        + sin(delta_c)^2*(1 -
area_ratio_term*cos(delta_c)^2) );
            CA_cont = max(CA_cont, eps);


            if Kn <= 1e-3
                CA = CA_cont;
            else
                s = V_eval/sqrt(2*R_air*Tinf);
                Tw_ratio_term = sqrt(pi*Tw/Tinf);


                CD_FM_sphere = 2 + (2/(3*s))*Tw_ratio_term;
                CD_FM_cone = sin(delta_c)^2 +
(1/s)*Tw_ratio_term*sin(delta_c) ...
                    + 2*cos(delta_c)^2;
                CA_FM = area_ratio_term*CD_FM_sphere + (1-
area_ratio_term)*CD_FM_cone;


                if Kn >= 10
                    CA = CA_FM;
                else
                    phi = pi*(3/8 + (1/8)*log10(Kn));
                    CA = CA_cont + (CA_FM - CA_cont)*sin(phi)^2;
                end
            end


            CD_table(i,j) = max(CA, eps);
```

```
            points_done = points_done + 1;


            pct_done = 100*points_done/total_points;
            if pct_done >= next_progress_pct
                elapsed = toc(build_tic);
                est_total = elapsed/points_done*total_points;
                fprintf('  %.0f%% done, elapsed %.1f min, est. total %.1f
min\n', pct_done, elapsed/60, est_total/60);
                next_progress_pct = next_progress_pct + 10;
            end
        end
    end
    fprintf('CD(V,h) table build complete in %.1f min.\n',
toc(build_tic)/60);


    CD_interp = griddedInterpolant({V_grid, h_grid}, CD_table, 'linear',
'none');
end


function xdot = EOM_6DOF(t, x, params)
% Non-planar EOM, Earth-fixed rotating frame (report Eq. 8-13). State:
% [V; gamma; h; psi; phi; theta]. gamma-cutoff termination (see
% run_dispersed_trajectory) avoids the 1/cos(gamma) singularity near
% vertical flight path angle.


    V     = x(1);
    gamma = x(2);
    h     = x(3);
    psi   = x(4);
    phi   = x(5);


    if V <= 0
        error('Nonpositive velocity at t=%.6f s.', t);
    end


    h_env = max(h, 0);
    Re    = params.Re;
    beta  = params.beta(V, h_env);
    LD    = params.LD(V, h_env);
    sigma = params.sigma(t);
    g     = params.g_fun(h_env);
    rho   = params.rho_fun(h_env);
```

```
    r     = Re + h;
    omega = params.omega;


    dVdt = -rho*V^2/(2*beta) + g*sin(gamma) ...
        - omega^2*r*cos(phi)*(sin(gamma)*cos(phi) +
cos(gamma)*sin(phi)*sin(psi));


    dgammadt = -(rho*V/(2*beta))*LD*cos(sigma) + g*cos(gamma)/V -
V*cos(gamma)/r ...
        - 2*omega*cos(phi)*cos(psi) ...
        - (omega^2*r/V)*cos(phi)*(cos(gamma)*cos(phi) -
sin(gamma)*sin(phi)*sin(psi));


    dhdt = -V*sin(gamma);


    dpsidt = (rho*V/(2*beta))*LD*sin(sigma)/cos(gamma) ...
        - V*cos(gamma)*cos(psi)*tan(phi)/r ...
        - 2*omega*(tan(gamma)*cos(phi)*sin(psi) + sin(phi)) ...
        - (omega^2*r/(V*cos(gamma)))*sin(phi)*cos(phi)*cos(psi);


    dphidt   = (V/r)*cos(gamma)*sin(psi);
    dthetadt = (V/r)*cos(gamma)*cos(psi)/cos(phi);


    xdot = [dVdt; dgammadt; dhdt; dpsidt; dphidt; dthetadt];
end
```

H. Mars Pathfinder Simulation Script

```
%% AE 6355 - Individual Project - Mars Tauber-Sutton Validation
(Pathfinder)


% Pathfinder is used because its entry velocity (7.48 km/s) is the one
% Mars entry in the NASA Probe Blue Book (Planetary Mission Entry
% Vehicles Quick Reference Guide v4.1) that falls inside Tauber-Sutton's
% valid Mars range (6.5-9 km/s); most Mars landers (Viking, MER,
% Phoenix, MSL, InSight, Mars2020) enter at 4.4-5.8 km/s, too slow for
% this correlation.
%
% Reference (Blue Book): peak convective ~100.69 W/cm^2, peak radiative
% ~5.27 W/cm^2 (with ablation).
```

```matlab
clear; clc; close all;


%% -------------------------------------------------------------------
%% MARS CONSTANTS (literature-sourced)
%% -------------------------------------------------------------------
g0_mars   = 3.71;        % m/s^2
Re_mars   = 3389.5e3;    % m
rho0_mars = 0.020;       % kg/m^3
H_mars    = 11.1e3;      % m


%% -------------------------------------------------------------------
%% PATHFINDER ENTRY CONDITIONS (Blue Book, p.16)
%% -------------------------------------------------------------------
V0     = 7480;               % m/s, relative entry velocity
gamma0 = deg2rad(14.06);     % rad, positive-down (sign-flipped from Blue
Book's -14.06 deg)
h0     = 130.86e3;           % m, entry interface altitude


rn         = 0.66;           % m, nose radius
beta_const = 61.5;           % kg/m^2, at peak dynamic pressure (Blue Book)
LD_const   = 0;              % ballistic


skip_altitude = h0 + 100;
tspan = [0 2000];
ode_settings = {'RelTol',1e-8, 'AbsTol',[1e-3 1e-10 1e-3], 'MaxStep',0.5};


%% -------------------------------------------------------------------
%% INTEGRATION
%% -------------------------------------------------------------------
params.Re = Re_mars;
params.beta = @(V,h) beta_const;
params.LD   = @(V,h) LD_const;
params.sigma = @(t) 0;
params.g_fun = @(h) g0_mars*(Re_mars/(Re_mars+max(h,0)))^2;
params.rho_fun = @(h) rho0_mars*exp(-max(h,0)/H_mars);


x0 = [V0; gamma0; h0];
opts = odeset(ode_settings{:}, 'Events', @(t,x)
ground_skip_event(t,x,skip_altitude));
[t, x, te, xe, ie] = ode45(@(t,x) EOM_3DOF(t, x, params), tspan, x0,
opts);
```

```matlab
if isempty(ie)
    outcome = 'time limit';
elseif ie(end) == 1
    outcome = 'ground impact';
else
    outcome = 'atmospheric skip-out';
end


V = x(:,1);
h = x(:,3);
fprintf('Outcome: %s at t=%.2f s (V=%.1f m/s, h=%.2f km)\n', outcome, te,
xe(1), xe(3)/1000);


%% ----------------------------------------------------------------
%% POST-HOC: HEATING
%% ----------------------------------------------------------------
rho_t = rho0_mars*exp(-max(h,0)/H_mars);


% Sutton-Graves convective heating. Derived for Earth air; applying it
% directly to Mars (CO2-dominated) is a known first-order approximation
% common in mission studies, not independently validated for Mars here.
k_SG = 1.7415e-4;
qdot_conv = k_SG*sqrt(rho_t/rn).*V.^3 / 1e4;   % W/cm^2
[qdot_conv_peak, idx_conv_peak] = max(qdot_conv);


% Mars Tauber-Sutton radiative heating (Eq. 28, Mars
% coefficients). Pathfinder's rn=0.66m falls below the correlation's
% stated 1-23m range; mild extrapolation, same caveat as Earth's rn.
C_M = 2.35e4;
a_M = 0.526;
b_M = 1.19;


V_kms = V/1000;
in_range_TS = V_kms >= 6.5 & V_kms <= 9.0;
qdot_rad = zeros(size(V));
qdot_rad(in_range_TS) = C_M * rn^a_M * rho_t(in_range_TS).^b_M ...
    .* tauber_sutton_fV(V(in_range_TS), 'mars');   % W/cm^2
[qdot_rad_peak, idx_rad_peak] = max(qdot_rad);
Q_rad_total = trapz(t, qdot_rad);


fprintf('\n--- MARS TAUBER-SUTTON VALIDATION (Pathfinder) ---\n');
```

```
fprintf('Peak convective (Sutton-Graves): %.2f W/cm^2 at t=%.1f s (V=%.2f
km/s)\n', ...
    qdot_conv_peak, t(idx_conv_peak), V(idx_conv_peak)/1000);
fprintf('  Reference (Blue Book): ~100.69 W/cm^2\n');
fprintf('Peak radiative (Tauber-Sutton, Mars): %.2f W/cm^2 at t=%.1f s
(V=%.2f km/s)\n', ...
    qdot_rad_peak, t(idx_rad_peak), V(idx_rad_peak)/1000);
fprintf('  Reference (Blue Book, with ablation): ~5.27 W/cm^2\n');
fprintf('Integrated radiative heat load: %.2f J/cm^2\n', Q_rad_total);


figure;
subplot(2,1,1);
plot(t, h/1000); xlabel('Time (s)'); ylabel('h (km)'); grid on;
title('Pathfinder Trajectory (Mars)');
subplot(2,1,2);
plot(t, qdot_conv, t, qdot_rad);
legend('Convective (Sutton-Graves)', 'Radiative (Tauber-Sutton, Mars)',
'Location','best');
xlabel('Time (s)'); ylabel('Heat rate (W/cm^2)'); grid on;


%% ====================================================================
%% LOCAL FUNCTIONS
%% ====================================================================


function [value, isterminal, direction] = ground_skip_event(t, x,
skip_altitude)
    h = x(3);
    value      = [h; h - skip_altitude];
    isterminal = [1; 1];
    direction  = [-1; +1];
end


function xdot = EOM_3DOF(t, x, params)
    V     = x(1);
    gamma = x(2);
    h     = x(3);


    if V <= 0
        error('Nonpositive velocity at t=%.6f s.', t);
    end


    h_env = max(h, 0);
```

```
    Re    = params.Re;
    beta  = params.beta(V, h_env);
    LD    = params.LD(V, h_env);
    sigma = params.sigma(t);
    g     = params.g_fun(h_env);
    rho   = params.rho_fun(h_env);


    dVdt     = -rho*V^2/(2*beta) + g*sin(gamma);
    dgammadt = -V*cos(gamma)/(Re + h) - (rho*V/(2*beta))*LD*cos(sigma) +
g*cos(gamma)/V;
    dhdt     = -V*sin(gamma);


    xdot = [dVdt; dgammadt; dhdt];
end


function fV = tauber_sutton_fV(V, planet)
% Tabulated Tauber-Sutton f(V), Table 1 (Tauber & Sutton, 1991), V in
% m/s. Log-space pchip interpolation.
    switch planet
        case 'earth'
            V_table = [9000,9250,9500,9750,10000,10250,10500,10750,11000,
...

11500,12000,12500,13000,13500,14000,14500,15000,15500,16000];
            fV_table =
[1.5,4.3,9.7,19.5,35,55,81,115,151,238,359,495,660,850,1065,1313,1550,1780
,2040];
        case 'mars'
            V_table =
[6000,6150,6300,6500,6700,6900,7000,7200,7400,7600,7800,8000,8200,8400,860
0,8800,9000];
            fV_table =
[0.2,1.0,1.95,3.42,5.1,7.1,8.1,10.2,12.5,14.8,17.1,19.2,21.4,24.1,26.0,28.
9,32.8];
        otherwise
            error('Unknown planet for tauber_sutton_fV: %s', planet);
    end


    fV = exp(interp1(V_table, log(fV_table), V, 'pchip', 'extrap'));
end
```

Acknowledgments

The author gratefully acknowledges the instructor and teaching staff of AE 6355 for their guidance throughout this project, including feedback on modeling assumptions, EquiFlow real-gas solver usage, and validation methodology against the Stardust's BET. The author also thanks Desai and Qualls for the detailed Stardust flight reconstruction data that made independent validation of this simulation possible, and the developers of EquiFlow for access to the equilibrium real-gas normal-shock and boundary-layer property solver used throughout this work.

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